



**SSN College Of
Engineering Kalavakkam-
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**Department of Computer Science and
Engineering UCS2404 Database Management
Systems Assignment 1**

Gas Supply Management System

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Problem Statement:

Analyse the data for a specific real-time application which uses a Gas supply chain Management system. Analysing the data includes identifying the constraints among the attributes and the existence of dependencies among the attributes. The constraints and dependencies identification could be framed/identified by deeper understanding of the problem specification, interactions/dependencies among various attributes and user-requirements for the specific real-time application. Prepare the documentation showing the list of attributes and identify various functional dependencies among the set of attributes.

Entities and Attributes:

1.Customer:

Attributes:

- cust_ID
- cust_name
- phone
- address
- password
- email
- agency_ID

2.Supplier:

Attributes:

- supplier_ID
- supplier_name
- phone
- address
- agency_ID
- supply_ID
- supply_date
- supply_qty
- gas_ID
- gas_type
- cost_per_unit

3.Agency:

Attributes:

- agency_ID
- agency_name
- phone
- address
- inventory_ID

- gas_id
- qty_avail

4.Gas_Booking:

Attributes:

- book_id
- cust_ID
- gas_ID
- qty_req
- book_date

5.Payment:

Attributes:

- pay_Id
- book_ID
- total_price
- pay_date

6.Delivery:

Attributes:

- delv_Id
- book_ID
- delv_date

7.Feedback:

Attributes:

- feedback_Id
- cust_Id
- feedback_content
- feedback_date

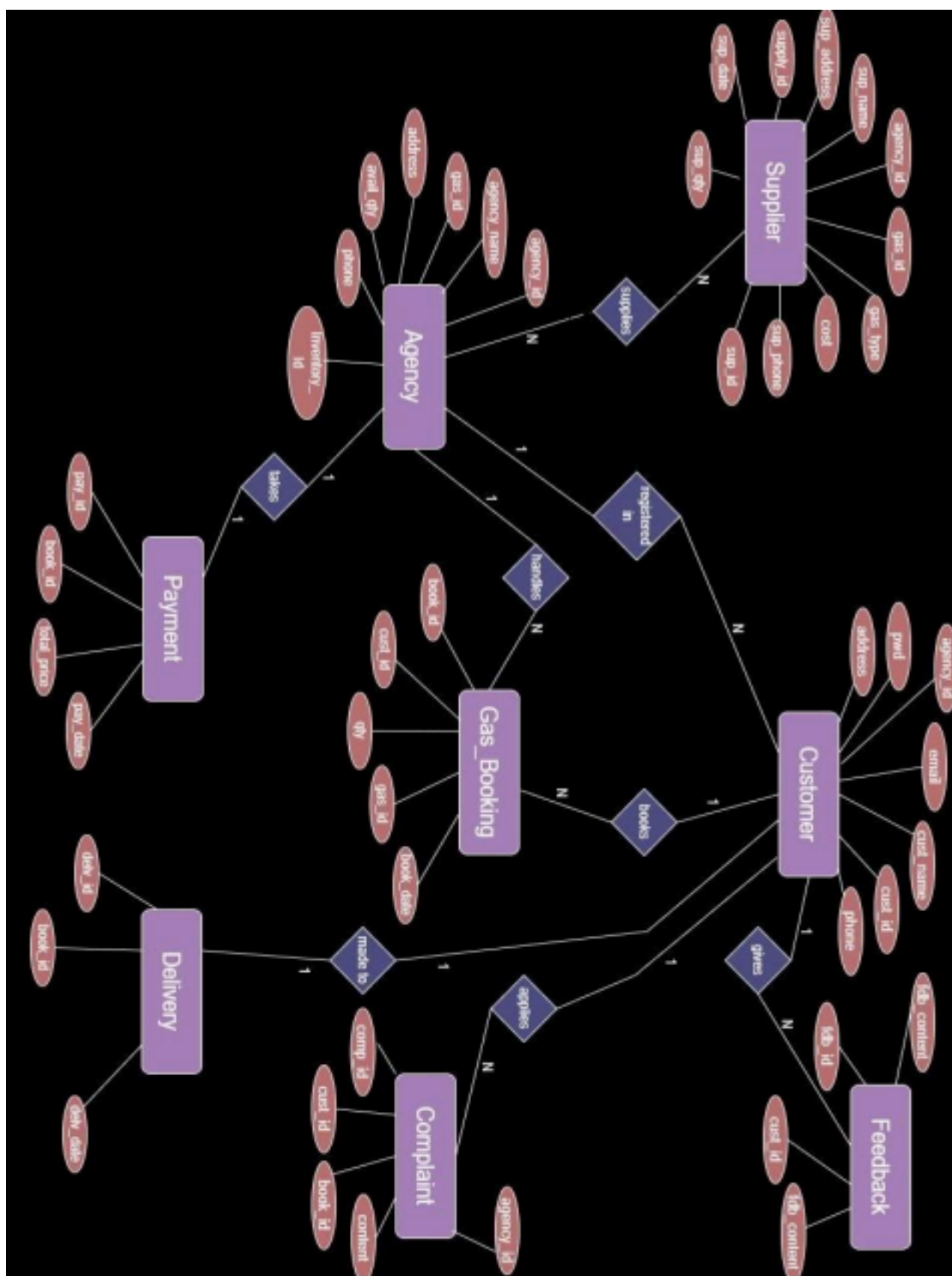
8.Complaint:

Attributes:

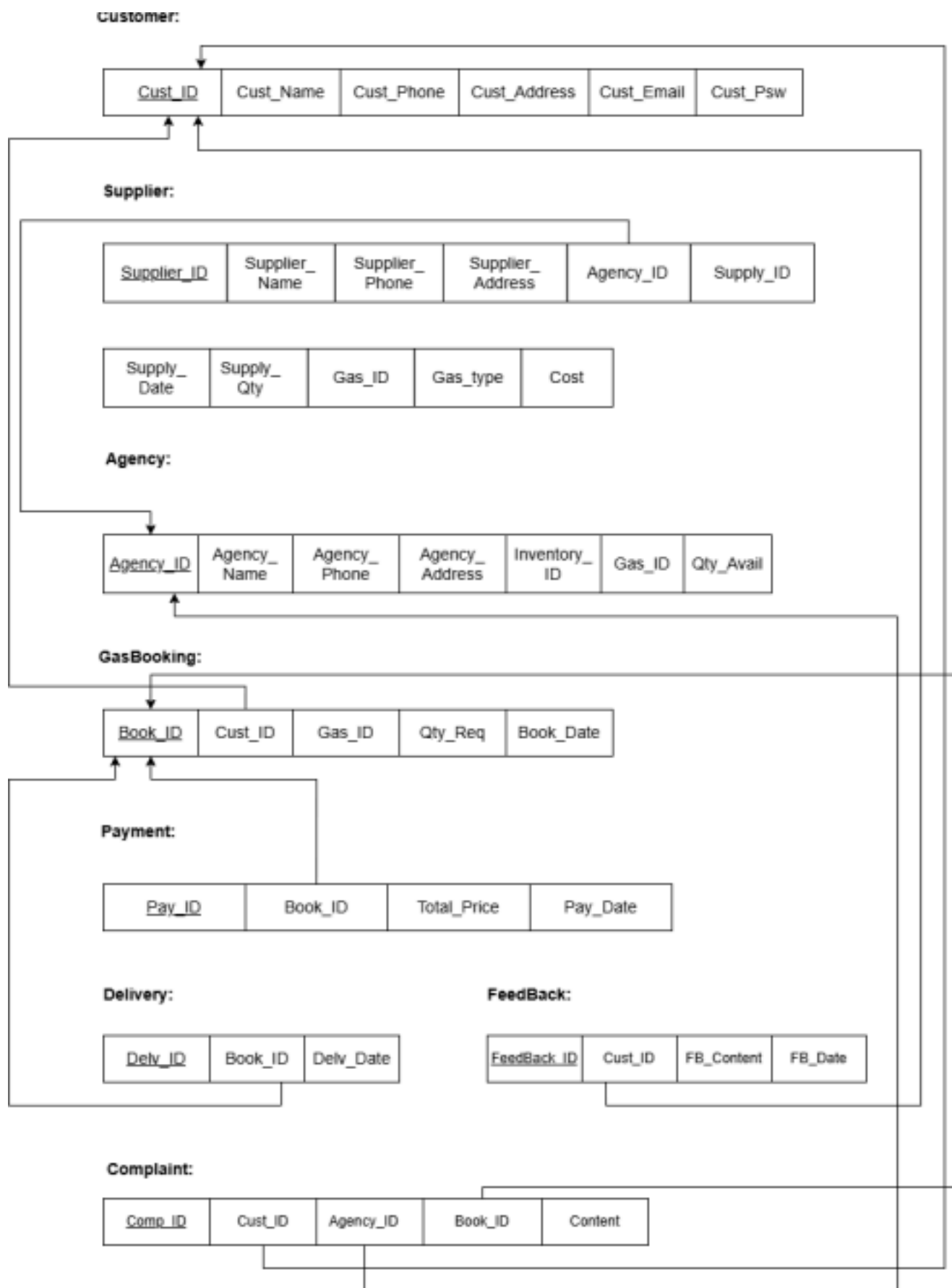
- comp_ID -
cust_ID -
agency_ID -
book_ID -
content

ER

Diagram:



Initial Schema:



Normalisation

1.Customer:

Attributes:

cust_ID -> A

cust_name -> B

phone -> C

address -> D

email -> E

password -> F

agency_ID -> G

$R = \{A, B, C, D, E, F, G\}$

Functional

Dependencies: A ->

BCDEFG

E -> F

Candidate Key:

A

Prime

Attributes(PA): A

Non Prime Attributes(NPA):

B,C,D,E,F,G

Normalisation:

Checking for 1NF:

Constraint: There are only unique values, no multivalued attributes. Automatically in 1NF.

Checking for 2NF:

Constraint: should be in 1NF and not have any partial dependencies.

- i) $A \rightarrow BCDEFG$
not partial dependencies.
- ii) $E \rightarrow F$
not a partial dependencies.

Therefore it is in 2nf.

Checking for 3NF:

Constraint: should be in 2NF and not have any transitive dependencies.(i.e. $NPA \rightarrow NPA$)

- i) $A \rightarrow BCDEFG$
It is a transitive dependency.

Decomposition:

$R1: \{E, F\}$

Candidate key: E

Prime Attribute: E

Non Prime attribute: F

$E \rightarrow F$ is not a transitive attribute

Thus, it is in 3nf.

R2: {A $\rightarrow$ BCDEG}

Candidate key: A

Prime Attribute: A

Non Prime Attributes: BCDEG

not a transitive attribute.

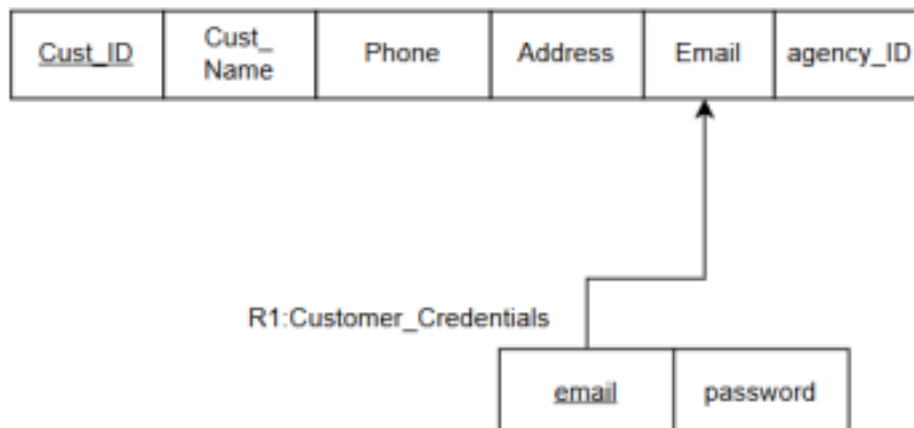
Thus, it is in 3nf.

ii) E $\rightarrow$ F

It is not a transitive dependency.

Decomposition Table:

R2:Customer_details



2.Supplier:

Attributes:

supplier_ID $\rightarrow$ A

supplier_name $\rightarrow$ B

phone $\rightarrow$ C

address -> D

agency_ID -> E

supply_ID -> F

supply_Date -> G

supply_Qty -> H

gas_ID -> I

gas_type -> J

cost_per_unit -> K

$R = \{A, B, C, D, E, F, G, H, I, J, K\}$

Functional

Dependencies: A ->

BCDE

F -> GHIJK

I -> JK

FQ -> H

Candidate Key:

AF

Prime

Attributes(PA): A,F

Non Prime

Attributes(NPA):

B,C,D,E,G,H,I,J,K

Normalisation:

Checking for 1NF:

Constraint: There are only unique values, no multivalued attributes. Automatically in 1NF.

Checking for 2NF:

Constraint: should be in 1NF and not have any partial dependencies.

i) $A \rightarrow BCDE$

It violates since it is a partial dependency

ii) $F \rightarrow GHIJK$

It violates since it is a partial dependency

iii) $I \rightarrow JK$

It is not a partial dependency.

iv) $FI \rightarrow H$

It is not a partial dependency.

Here 2 FDs violate 2nf rule.

Decomposition:

R1: {A,B,C,D,E}

FDs: { $A \rightarrow BCDE$ }

Candidate Key: A

Prime Attribute: A

Non Prime Attributes: BCDE

Now this relation is in 2nf.

R2: {F,G,H,I,J,K}

FDs: {F→GHIJK, I→JK, FI→H}

Candidate key: F

Prime attributes: F

Non Prime Attributes: GHIJK

Now this relation is in 2nf.

R3: {A,F}

FDs: {}

Now this relation is in 2nf.

Checking for 3NF:

Constraint: should be in 2NF and not have any transitive dependencies.(i.e. NPA → NPA)

i) R1: { ABCDE }

It is not a transitive dependency.

ii) R2: {FGHIJK}

It violates since it's a transitive dependency.

iii) R3: {A,F}

It is not a transitive dependency.

Decomposition:

R2: {FGHIJK}

R21: {IJK}

FDs: {I->JK}

Candidate key: I

Prime Attributes: I

Non Prime Attributes :

J,K Now this relation is in

3nf.

R22: {FIH}

FDs: { FI->H}

Candidate key: FI

Prime attributes : F,I

Non Prime Attributes: J,K

Now this relation is in

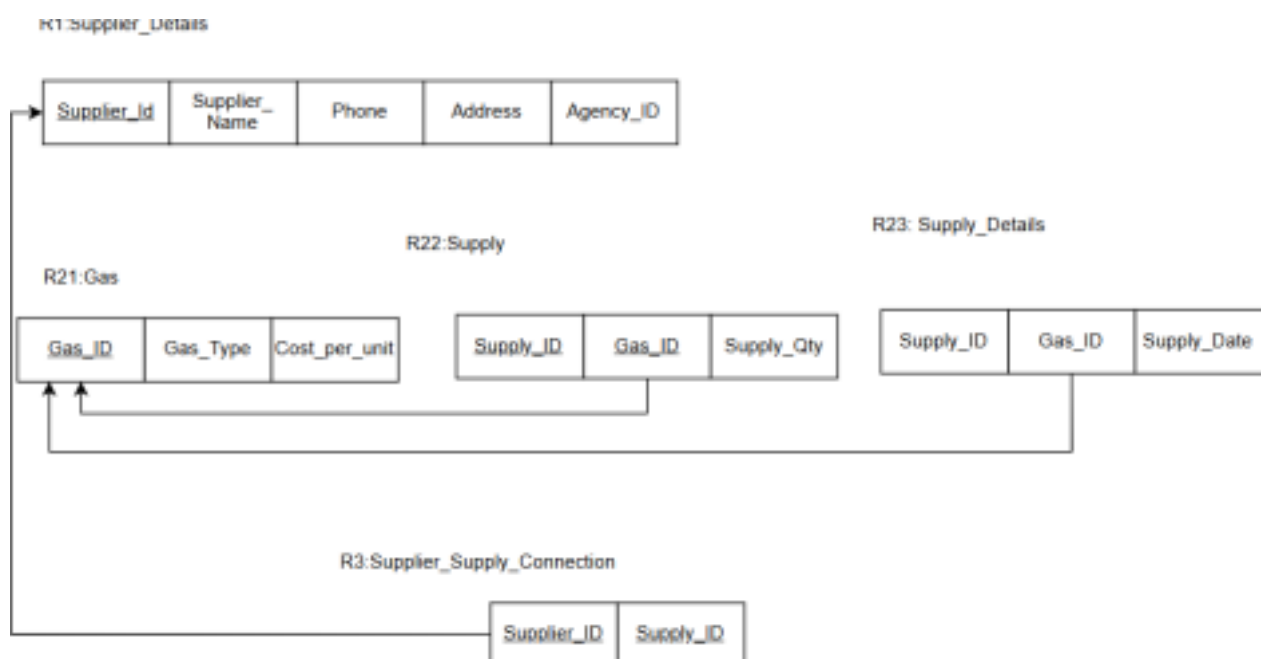
3nf.

R23: {F,I,G}

FDs: {F->GI }

This relation is in 2nf and

3nf. Decomposition Table:



3. Agency:

Attributes:

agency_ID -> A

agency_name -> B

phone -> C

address $\rightarrow$ D

inventory_ID $\rightarrow$ E

gas_Id $\rightarrow$ F

qty_avail $\rightarrow$ G

$R = \{A, B, C, D, E, F, G\}$

Functional Dependencies:

$A \rightarrow BCD$

$AE \rightarrow FG$

Candidate Key:

AE

Prime Attributes(PA):

A, E

Non Prime Attributes(NPA):

B, C, D, F, G

Normalisation:

Checking for 1NF:

Constraint: There are only unique values, no multivalued attributes. Automatically in 1NF.

Checking for 2NF:

Constraint: should be in 1NF and not have any partial dependencies.

i) $A \rightarrow BCD$

It violates since it's a partial dependency.

Decomposition:

$R1: \{ABCD\}$

$FDs: \{A \rightarrow BCD\}$

Candidate Key: A

Prime Attributes: A

Non Prime Attributes: B,C,D

Now this is in 2nf because no partial dependency.

$R2: \{AEFG\}$

$FDs: \{AE \rightarrow FG\}$

Candidate key: AE

Prime Attributes: A,E

Non Prime Attributes : F,G

Now this is in 2nf.

ii) $AE \rightarrow FG$

It is not a partial dependency.

Checking for 3NF:

Constraint: should be in 2NF and not have any transitive dependencies.(i.e. $NPA \rightarrow NPA$)

i) $R1: \{ABCD\}$

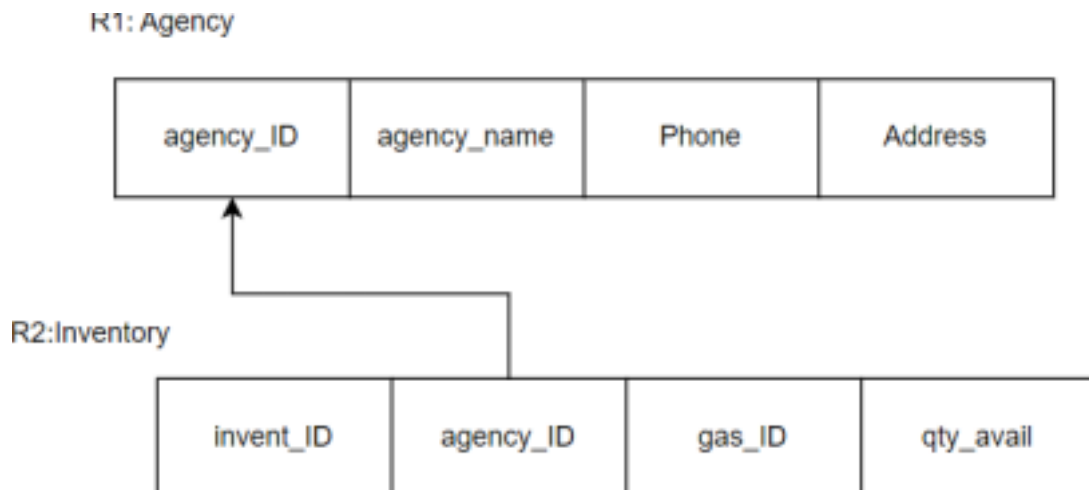
It is not a transitive dependency.

ii) $R2: \{AEFG\}$

It is not a transitive dependency

It is in 3nf .

Decomposition table:



4.Gas_Booking:

Attributes:

book_ID -> A

cust_ID -> B

gas_ID -> C

qty_req -> D

book_date->E

$R=\{A,B,C,D,E\}$

Functional Dependencies:

$A \rightarrow BCDE$

Candidate Key:

A

Prime Attributes(PA):

A

Non Prime Attributes(NPA):

B,C,D,E

Normalisation:

Checking for 1NF:

Constraint: There are only unique values, no multivalued attributes. Automatically in 1NF.

Checking for 2NF:

Constraint: should be in 1NF and not have any partial dependencies.

$A \rightarrow BCDE$

It is not a partial dependency.

It is in 2nf.

Checking for 3NF:

Constraint: should be in 2NF and not have any transitive dependencies.(i.e. $NPA \rightarrow NPA$)

$A \rightarrow BCD$

It is not a transitive dependency.

It is in 3nf.

5.Payment:

Attributes:

pay_ID -> A

book_ID -> B

total_price -> C

pay_date -> D

$R = \{A, B, C, D\}$

Functional Dependencies:

$A \rightarrow BCD$

$B \rightarrow CD$

removing $B \rightarrow CD$ because redundancy

Candidate Key:

A

Prime Attributes(PA):

A

Non Prime Attributes(NPA):

B, C, D

Normalisation:

Checking for 1NF:

Constraint: There are only unique values, no multivalued attributes. Automatically in 1NF.

Checking for 2NF:

Constraint: should be in 1NF and not have any partial dependencies.

$A \rightarrow BCD$

It is not a partial dependency.

It is in 2nf.

Checking for 3NF:

Constraint: should be in 2NF and not have any transitive dependencies.(i.e. $NPA \rightarrow NPA$)

$A \rightarrow BCD$

It is not a transitive dependency.

It is in 3nf.

6.Delivery:

Attributes:

$delv_ID \rightarrow A$

$book_ID \rightarrow B$

$delv_date \rightarrow C$

$R = \{A, B, C\}$

Functional Dependencies:

$$A \rightarrow BC$$
Candidate Key:

$$A$$
Prime Attributes(PA):

$$A$$
Non Prime Attributes(NPA):

$$B, C$$
Normalisation:**Checking for 1NF:**

Constraint: There are only unique values, no multivalued attributes. Automatically in 1NF.

Checking for 2NF:

Constraint: should be in 1NF and not have any partial dependencies.

$$A \rightarrow BC$$

It is not a partial dependency.

It is in 2nf.

Checking for 3NF:

Constraint: should be in 2NF and not have any transitive

dependencies.(i.e. NPA $\rightarrow$ NPA)

$A \rightarrow BC$

It is not a transitive dependency.

It is in 3nf.

7.Feedback:

Attributes:

feedback_ID $\rightarrow$ A

cust_ID $\rightarrow$ B

feedback_content $\rightarrow$ C

feedback_date $\rightarrow$ D

$R=\{A,B,C,D\}$

Functional Dependencies:

$A \rightarrow BCD$

Candidate Key:

A

Prime Attributes(PA):

A

Non Prime Attributes(NPA):

B,C,D

Normalisation:

Checking for 1NF:

Constraint: There are only unique values, no multivalued attributes. Automatically in 1NF.

Checking for 2NF:

Constraint: should be in 1NF and not have any partial dependencies.

$A \rightarrow BCD$

It is not a partial dependency.

It is in 2nf.

Checking for 3NF:

Constraint: should be in 2NF and not have any transitive dependencies.(i.e. $NPA \rightarrow NPA$)

$A \rightarrow BCD$

It is not a transitive dependency.

It is in 3nf.

8.Complaint:

Attributes:

comp_ID $\rightarrow$ A

cust_ID $\rightarrow$ B

agency_ID $\rightarrow$ C

book_ID $\rightarrow$ D

content $\rightarrow$ E

$R = \{A, B, C, D, E\}$

Functional

Dependencies: A $\rightarrow$

BCDE

B $\rightarrow$ C

Candidate Key:

A

Prime

Attributes(PA): A

Non Prime

Attributes(NPA): B, C, D, E

Normalisation:

Checking for 1NF:

Constraint: There are only unique values, no multivalued attributes. Automatically in 1NF.

Checking for 2NF:

Constraint: should be in 1NF and not have any partial dependencies. $A \rightarrow BCDE$

It is not a partial dependency.

$B \rightarrow C$

It is not a partial dependency.

It is in 2nf.

Checking for 3NF:

Constraint: should be in 2NF and not have any transitive dependencies.(i.e. $NPA \rightarrow NPA$)

$A \rightarrow BCDE$

It is not a transitive dependency.

$B \rightarrow C$

It is in 3nf.

It is not a transitive dependency.

Entities and Attributes After Normalization:

1. Customer_Credentials

- email
- password

2. Customer

- cust_ID
- cust_name
- phone
- address
- email
- Agency_ID

3. Supplier

- supplier_ID
- supplier_name
- phone
- address
- Agency_ID

4. Supplier_Supply_Connection

- supplier_id
- supply_id

5. Gas

- gas_id
- gas_type
- cost_per_unit

6. Supply

- Supply_ID
- Gas_id
- supply_qty

7. Supply_Details

- supply_ID

- gas_id
- Supply_date

8. Agency

- agency_id
- agency_name ·
- phone
- address

9. Inventory

- inventory_ID ·
- agency_id
- gas_id
- qty_avail

10. Gas_Booking

- book_id
- cust_id
- gas_id
- qty_req
- book_date

11. Payment

- pay_id
- book_id
- total_price
- pay_date

12. Delivery

- delv_id
- book_id
- delv_date

13. Feedback

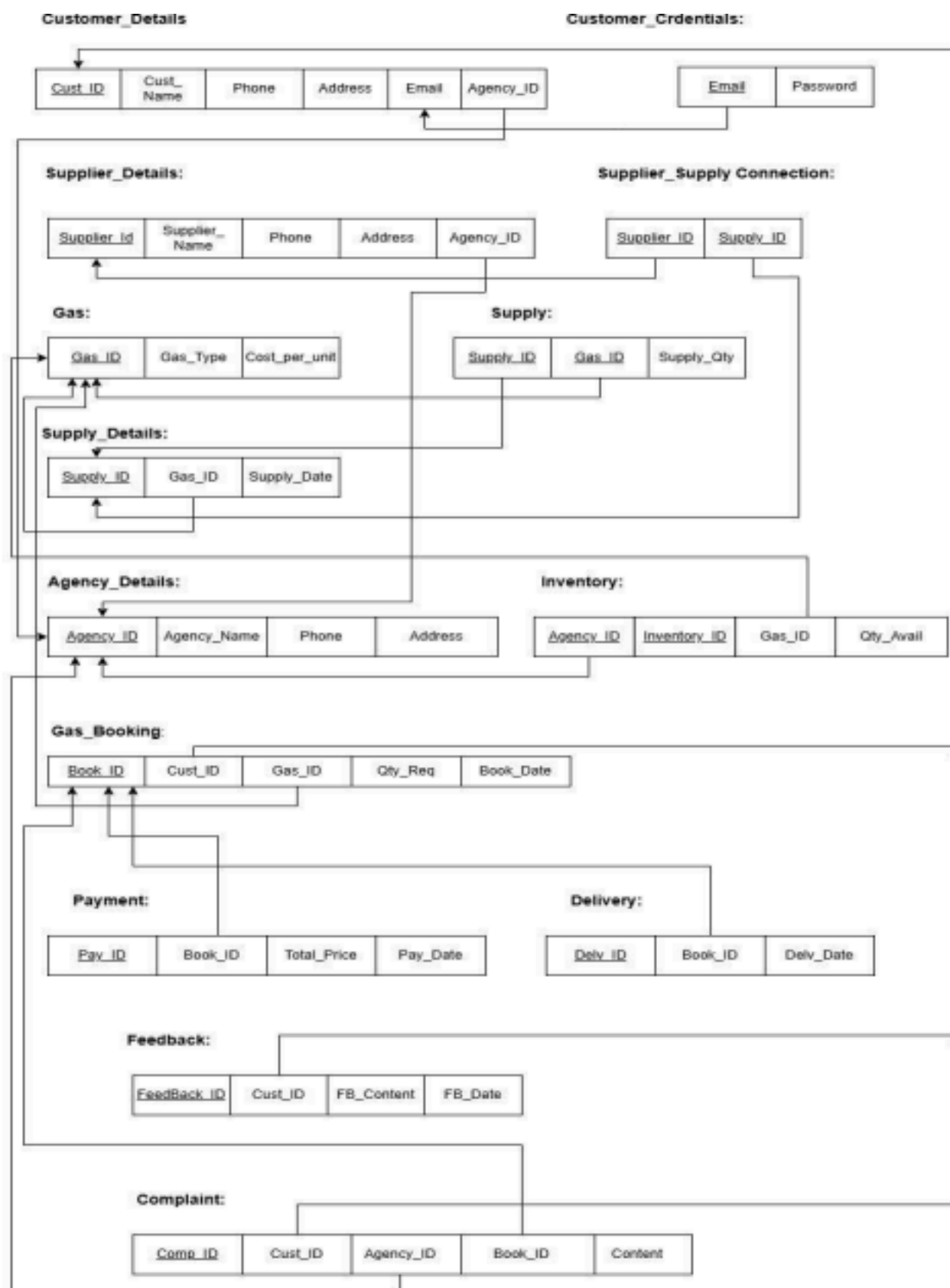
- feedback_id ·
- cust_ID
- FB_Content ·

FB_Date

14.Complaint

- comp_id
- cust_id
- agency_id •
- book_id
- content

Final Schema:



Assumptions and Work Flow:

1. Customer:

- We assume that each Customer is registered with only one agency.
- Customers have unique passwords and email for their login

2. Supplier:

- Suppliers supply stock to agencies
- Suppliers have their own contact details.

3. Agency and Inventory:

- Every agency has an Inventory which has the details of which gas is available and how much quantity of that is available(this stock is received from the Supplier).
- Each agency is connected with multiple Customers.

4. Gas Booking:

- Here we check if the qty required by customer is available at the inventory of the gas agency or not.
- If not we display a 'Out of Stock' message.

5. Transaction:

- After successful gas booking, the total price is displayed to the customer and payment is processed.

6. Delivery:

- delivery is made to the customer and the delivery date is noted

7. Customer Interactions with agency:

- a Customer can file complaints and give feedbacks regarding the agency and its delivery.

Creation of Tables

```
CREATE TABLE Customer_Credentials (
  email VARCHAR(50),
  password VARCHAR(20) UNIQUE,
  CONSTRAINT pk_email PRIMARY KEY (email),
  CONSTRAINT email_check CHECK (email LIKE '%@gmail.com')
);
```

```
CREATE TABLE Customer (
  cust_ID CHAR(5),
  cust_name VARCHAR2(25),
  phone NUMBER(10) UNIQUE,
  address VARCHAR(50),
  email VARCHAR(50),
  Agency_ID agency(4),
  CONSTRAINT fk_agency
  FOREIGN KEY (Agency_ID)
  REFERENCES Agency(Agency_ID),
  CONSTRAINT fk_customer_credentials
  FOREIGN KEY (email)
  REFERENCES Customer_Credentials(email),
  CONSTRAINT cust_id_check CHECK (cust_ID LIKE 'C%'),
  CONSTRAINT pk_cust_id PRIMARY KEY (cust_ID)
);
```

```
CREATE TABLE Supplier (
  supplier_ID CHAR(5),
  supplier_name VARCHAR(25),
  phone NUMBER(10) UNIQUE,
  address VARCHAR(50),
  Agency_ID char(4),
  CONSTRAINT pk_supplier PRIMARY KEY (supplier_ID),
  CONSTRAINT chk_supplier_id CHECK (supplier_ID LIKE 'S%'), CONSTRAINT
  fk_agency_supplier FOREIGN KEY (Agency_ID) REFERENCES
  Agency(Agency_ID)
);
```

```

CREATE TABLE Supplier_Supply_Connection (
  supplier_id CHAR(5),
  supply_id CHAR(5),
  CONSTRAINT pk_supplier_supply_connection PRIMARY KEY (supplier_id,
  supply_id),
  CONSTRAINT fk_supplier_id FOREIGN KEY (supplier_id) REFERENCES
  Supplier(supplier_ID),
  CONSTRAINT fk_supply_id FOREIGN KEY (supply_id) REFERENCES
  Supply_Details(supply_ID)
);

```

```

CREATE TABLE Gas (
  gas_id CHAR(5) CONSTRAINT pk_gas PRIMARY KEY CHECK (gas_id LIKE
  'G%'),
  gas_type VARCHAR(25),
  cost_per_unit DECIMAL
);

```

```

CREATE TABLE Supply (
  Supply_ID CHAR(5),
  Gas_id CHAR(5),
  supply_qty NUMBER,
  CONSTRAINT pk_supply PRIMARY KEY (Supply_ID, Gas_id),
  CONSTRAINT fk_supply_id FOREIGN KEY (Supply_ID) REFERENCES
  Supply_Details(supply_ID),
  CONSTRAINT fk_gas_id FOREIGN KEY (Gas_id) REFERENCES Gas(gas_id)
);

```

```

CREATE TABLE Supply_details (
  supply_ID CHAR(5) CONSTRAINT pk_supply_details PRIMARY KEY CHECK
  (supply_ID LIKE 'SU%'),
  gas_id CHAR(5) REFERENCES Gas(gas_id),
  Supply_date DATE
);

```

```

CREATE TABLE Agency (

```

```

agency_id CHAR(4) CONSTRAINT pk_agency PRIMARY KEY CHECK (agency_id
LIKE 'A%'),
agency_name VARCHAR(15),

phone NUMBER(10),
address VARCHAR(25)
);

```

```

CREATE TABLE Inventory (
inventory_ID CHAR(4) CONSTRAINT pk_inventory PRIMARY KEY CHECK
(inventory_ID LIKE 'I%'),
agency_id CHAR(4),
gas_id CHAR(4) REFERENCES Gas(gas_id),
qty_avail NUMBER(50),
CONSTRAINT fk_agency_inventory FOREIGN KEY (agency_id) REFERENCES
Agency(agency_id)
);

```

```

CREATE TABLE Gas_Booking (
book_id CHAR(4) CONSTRAINT pk_gas_booking PRIMARY KEY CHECK
(book_id LIKE 'B%'),
cust_id CHAR(5) REFERENCES Customer(cust_ID),
gas_id CHAR(4) REFERENCES Gas(gas_id),
qty_req NUMBER(50),
book_date DATE
);

```

```

CREATE TABLE Payment (
pay_id CHAR(4) CONSTRAINT pk_payment PRIMARY KEY CHECK (pay_id
LIKE 'P%'),
book_id CHAR(4) REFERENCES Gas_Booking(book_id),
total_price DECIMAL,
pay_date DATE
);

```

```

CREATE TABLE Delivery (
delv_id CHAR(4) CONSTRAINT pk_delivery PRIMARY KEY CHECK (delv_id
LIKE 'D%'),
book_id CHAR(4) REFERENCES Gas_Booking(book_id),
delv_date DATE

```

);

CREATE TABLE Feedback (

feedback_id CHAR(4) CONSTRAINT pk_feedback PRIMARY KEY CHECK
(feedback_id LIKE 'F%'),
cust_ID CHAR(5) REFERENCES Customer(cust_ID),
FB_Content VARCHAR(100),
FB_Date DATE
);

CREATE TABLE Complaint (

comp_id CHAR(4) CONSTRAINT pk_complaint PRIMARY KEY CHECK (comp_id
LIKE 'C%'),
cust_id CHAR(5) REFERENCES Customer(cust_ID),
agency_id CHAR(4) REFERENCES Agency(agency_id),
book_id CHAR(4) REFERENCES Gas_Booking(book_id),
content VARCHAR(100)
);

Triggers/Procedures used:

1. Trigger for Gas Booking to Check Inventory

This trigger will fire before an insert on the **Gas_Booking** table. It will check if the requested quantity is available in the inventory. If not, it will raise an exception.

```
CREATE OR REPLACE TRIGGER trg_check_inventory_before_booking
BEFORE INSERT ON Gas_Booking
FOR EACH ROW
DECLARE
    total_qty_avail NUMBER;
    v_agency_id CHAR(4);
```

```

BEGIN
    -- Retrieve the agency_id from the Customer table based on cust_id
    SELECT agency_id INTO v_agency_id
    FROM Customer
    WHERE cust_ID = :NEW.cust_id;

    -- Fetch the total available quantity from the inventory for the specific
    gas_id and agency_id
    SELECT SUM(qty_avail)
    INTO total_qty_avail
    FROM Inventory
    WHERE gas_id = :NEW.gas_id
      AND agency_id = v_agency_id;

    -- Check if the requested quantity is available
    IF :NEW.qty_req > total_qty_avail THEN
        RAISE_APPLICATION_ERROR(-20001, 'Out of stock');
    END IF;
END;
/

```

2. Trigger to Update Inventory after Payment

This trigger will fire after an insert on the `Payment` table. It will deduct the booked quantity from the inventory.

```

CREATE OR REPLACE TRIGGER trg_update_inventory_after_payment
AFTER INSERT ON Payment
FOR EACH ROW

```

DECLARE

v_qty_req NUMBER;

v_gas_id CHAR(4);

v_agency_id CHAR(4);

v_cust_id CHAR(5);

BEGIN

-- Fetch the requested quantity, gas_id, and cust_id from the Gas_Booking table

BEGIN

SELECT qty_req, gas_id, cust_id INTO v_qty_req, v_gas_id, v_cust_id

FROM Gas_Booking

WHERE book_id = :NEW.book_id;

EXCEPTION

WHEN NO_DATA_FOUND THEN

RAISE_APPLICATION_ERROR(-20001, 'No corresponding booking found in Gas_Booking table for book_id: ' || :NEW.book_id);

WHEN TOO_MANY_ROWS THEN

RAISE_APPLICATION_ERROR(-20002, 'Multiple bookings found in Gas_Booking table for book_id: ' || :NEW.book_id);

WHEN OTHERS THEN

RAISE_APPLICATION_ERROR(-20003, 'Error fetching data from Gas_Booking table: ' || SQLERRM);

END;

-- Fetch the agency_id from the Customer table based on cust_id

BEGIN

SELECT agency_id INTO v_agency_id

FROM Customer

WHERE cust_ID = v_cust_id;

EXCEPTION

WHEN NO_DATA_FOUND THEN

RAISE_APPLICATION_ERROR(-20005, 'No corresponding customer

```

found in Customer table for cust_id: ' || v_cust_id);
    WHEN TOO_MANY_ROWS THEN
        RAISE_APPLICATION_ERROR(-20006, 'Multiple customers found in
Customer table for cust_id: ' || v_cust_id);
    WHEN OTHERS THEN
        RAISE_APPLICATION_ERROR(-20007, 'Error fetching data from
Customer table: ' || SQLERRM);
END;

-- Update the inventory to deduct the booked quantity
BEGIN
    UPDATE Inventory
    SET qty_avail = qty_avail - v_qty_req
    WHERE gas_id = v_gas_id
    AND agency_id = v_agency_id;

    IF SQL%NOTFOUND THEN
        RAISE_APPLICATION_ERROR(-20004, 'No corresponding gas_id and
agency_id found in Inventory table: ' || v_gas_id || ', ' || v_agency_id);
    END IF;
EXCEPTION
    WHEN OTHERS THEN
        RAISE_APPLICATION_ERROR(-20008, 'Error updating Inventory table:
' || SQLERRM);
END;
END;
/

```

3. Trigger to Calculate Total Price in Payment

This trigger will fire before an insert on the `Payment` table. It will calculate the total price based on the requested quantity and the cost per unit of the gas and insert into the table `total_amount`.

```
CREATE OR REPLACE TRIGGER trg_calculate_total_amount
AFTER INSERT ON Gas_Booking
FOR EACH ROW
DECLARE
    v_cost_per_unit NUMBER(10, 2);
BEGIN
    -- Fetch cost per unit from Gas table based on gas_id
    SELECT cost_per_unit INTO v_cost_per_unit
    FROM Gas
    WHERE gas_id = :NEW.gas_id;

    -- Calculate total amount
    :NEW.total_amount := v_cost_per_unit * :NEW.qty_req;

    -- Insert or update total amount in Total_Amount table
    INSERT INTO Total_Amount (book_id, total_amount)
    VALUES (:NEW.book_id, :NEW.total_amount);
END;
/
```

Table Creation to store total amount:

```
CREATE TABLE Total_Amount (
    book_id CHAR(4) PRIMARY KEY,
    total_amount NUMBER(10, 2)
);
```

4. Trigger to check the booking quantity exceeds limit

This trigger will check the booking quantity. If it is greater than 3 , gas cannot be booked.

```
CREATE OR REPLACE TRIGGER trg_check_qty_req
```

```

2 BEFORE INSERT OR UPDATE ON Gas_Booking
3 FOR EACH ROW
4 BEGIN
5     IF :NEW.qty_req > 3 THEN
6         RAISE_APPLICATION_ERROR(-20001, 'Requested quantity exceeds the limit
of 3.');
```

```

7     END IF;
```

```

8 END;
```

```

9 /
```

5. Trigger to check the interval of the gas booking.

This trigger will check the interval between last booking and current booking. If it is less than 30 days, gas cannot be booked.

```

CREATE OR REPLACE TRIGGER trg_check_booking_interval
```

```

BEFORE INSERT ON Gas_Booking
```

```

FOR EACH ROW
```

```

DECLARE
```

```

    last_booking_date DATE;
```

```

BEGIN
```

```

    -- Retrieve the date of the last booking for the same customer
```

```

    SELECT MAX(book_date)
```

```

    INTO last_booking_date
```

```

    FROM Gas_Booking
```

```

    WHERE cust_id = :NEW.cust_id;
```

```

    -- Check if the difference between the last booking date and the new booking date is
greater than 30 days
```

```

    IF last_booking_date IS NOT NULL AND :NEW.book_date <= last_booking_date +
30 THEN
```

```

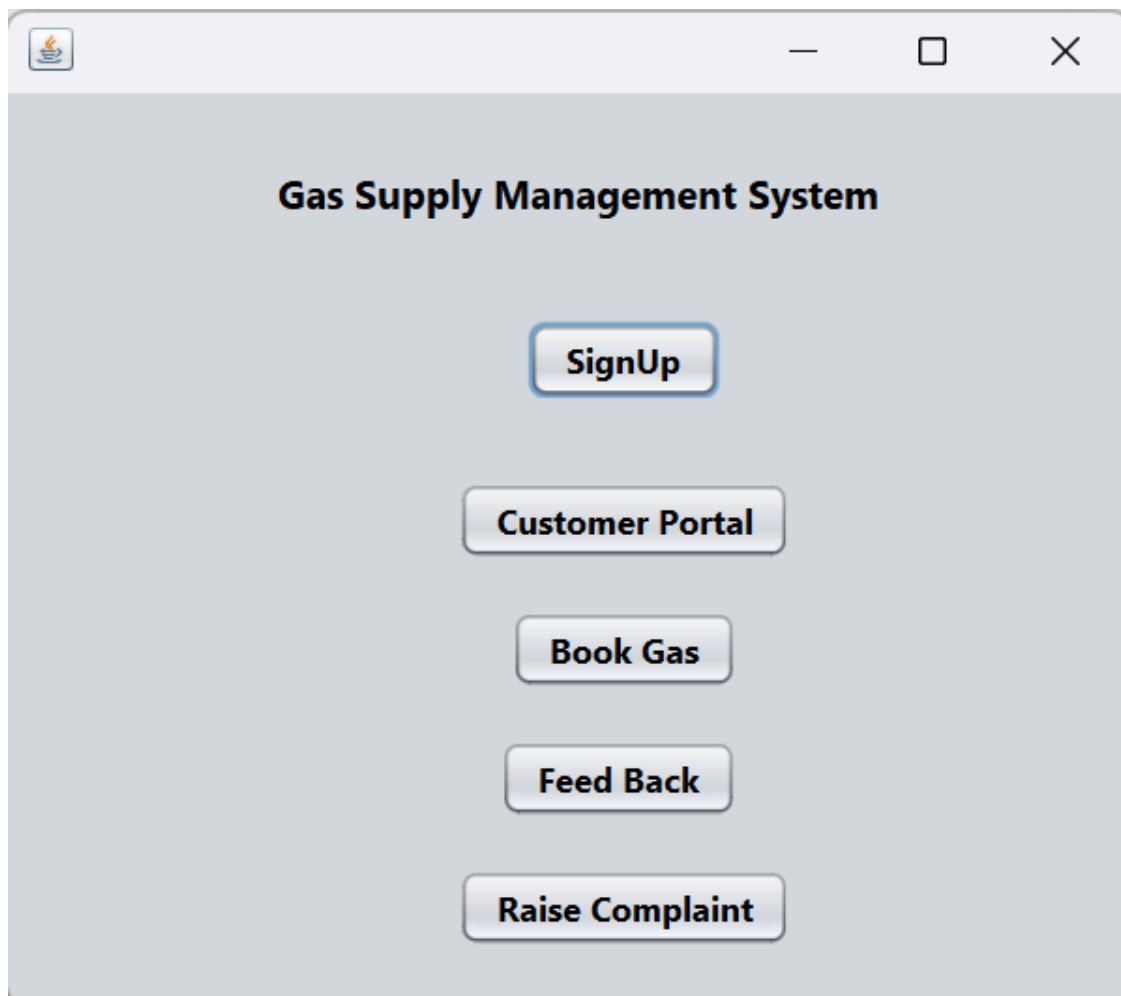
        RAISE_APPLICATION_ERROR(-20001, 'You cannot book gas within 30 days of
the last booking.');
```

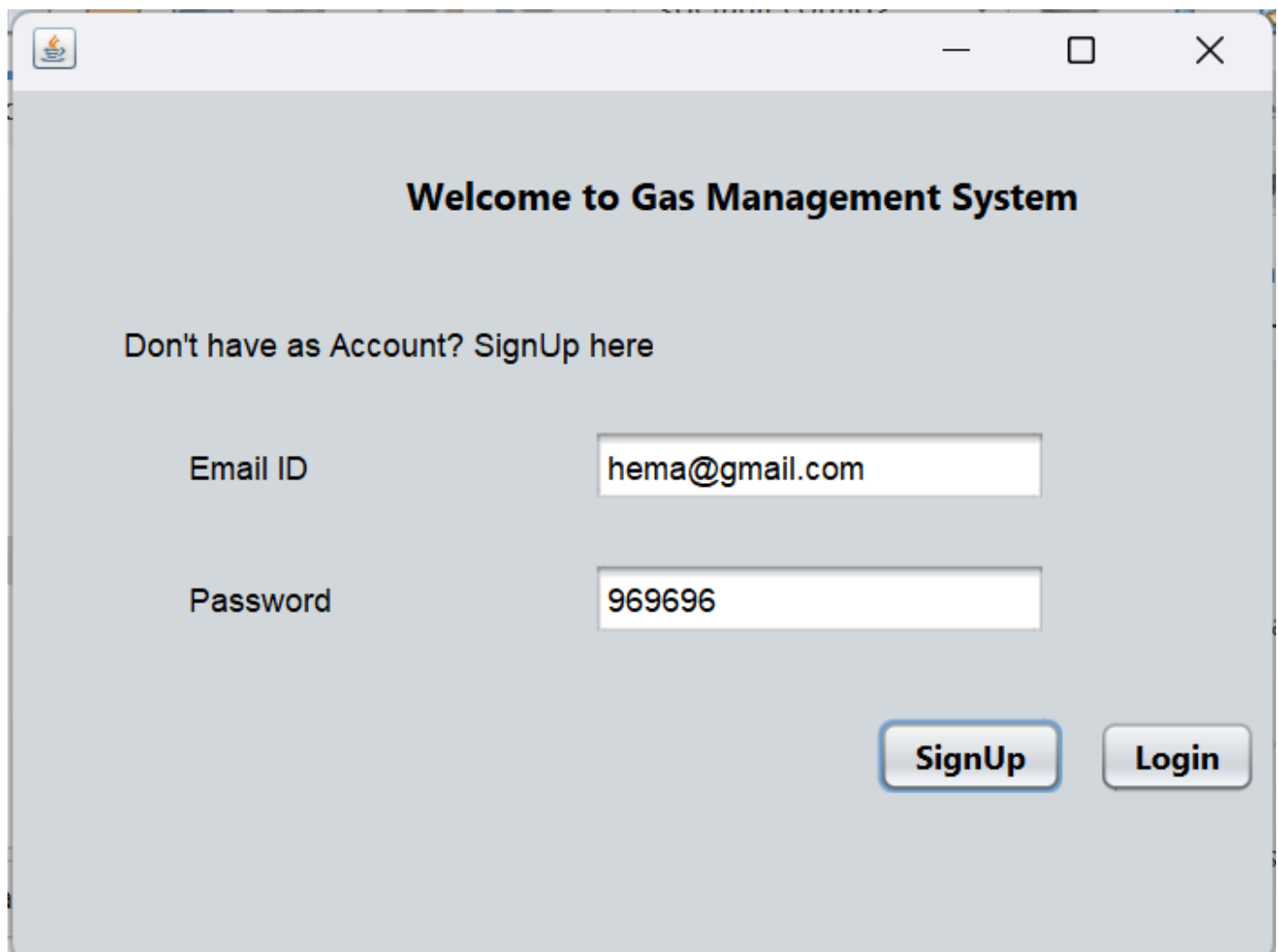
```

    END IF;
```

Screenshots of Work Flow:

Customer side:





A screenshot of a web application window titled "Welcome to Gas Management System". The window has a light gray background and a standard browser window header with a minimize, maximize, and close button. The main content area contains a heading "Welcome to Gas Management System" in bold black text. Below the heading is a link "Don't have as Account? SignUp here". There are two input fields: "Email ID" with the value "hema@gmail.com" and "Password" with the value "969696". At the bottom right, there are two buttons: "SignUp" and "Login".

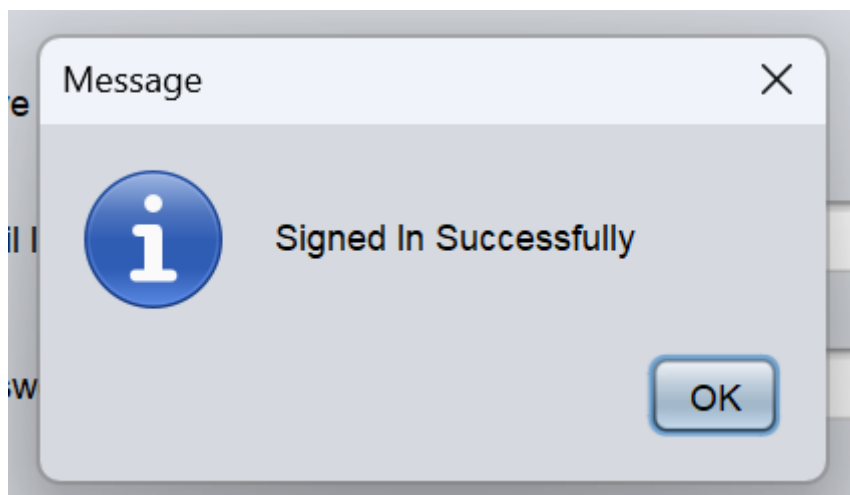
Welcome to Gas Management System

Don't have as Account? [SignUp here](#)

Email ID

Password

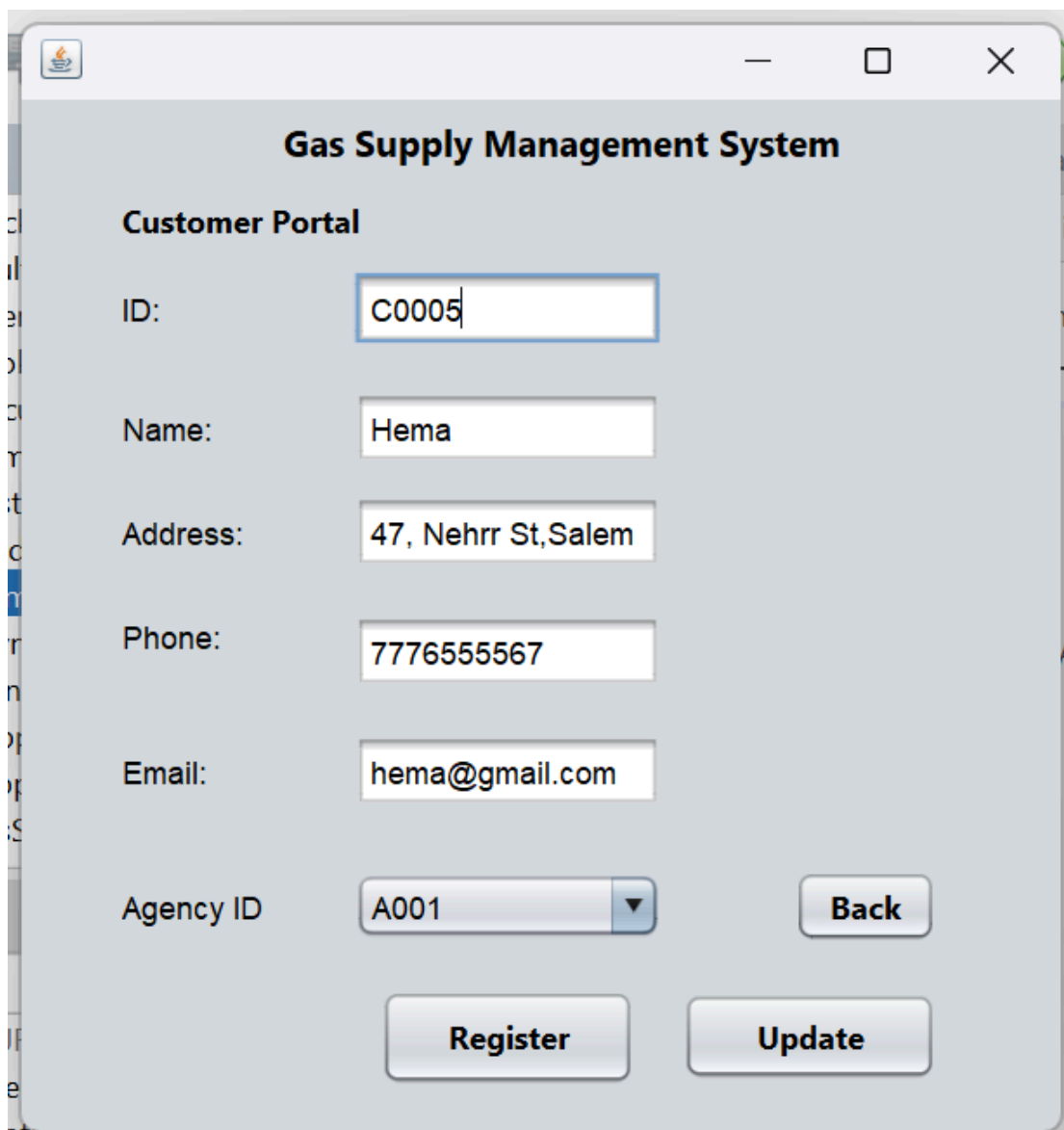
[SignUp](#) [Login](#)



EMAIL	PASSWORD
shanks@gmail.com	123456
raja@gmail.com	777000
shreya@gmail.com	434343
meena@gmail.com	444333
abc@gmail.com	345678
lmn@gmail.com	666000
hema@gmail.com	969696
7 rows selected.	

#	EMAIL	PASSWORD
1	shanks@gmail.com	123456
2	raja@gmail.com	777000
3	shreya@gmail.com	434343
4	meena@gmail.com	444333
5	abc@gmail.com	345678
6	lmn@gmail.com	666000
7	hema@gmail.com	969696

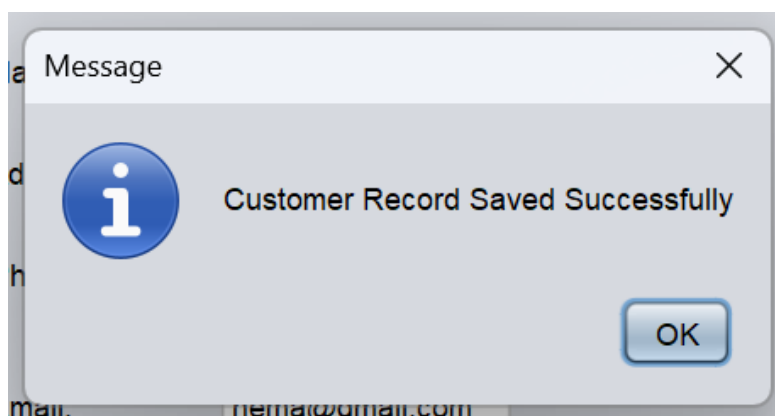
Note: Gmail and password is inserted in the customer credentials table.



The image shows a software window titled "Gas Supply Management System" with a "Customer Portal" section. It contains several input fields for customer information: ID (C0005), Name (Hema), Address (47, Nehrr St,Salem), Phone (7776555567), and Email (hema@gmail.com). There is also a dropdown menu for Agency ID set to A001. At the bottom, there are three buttons: "Back", "Register", and "Update".

Field	Value
ID:	C0005
Name:	Hema
Address:	47, Nehrr St,Salem
Phone:	7776555567
Email:	hema@gmail.com
Agency ID	A001

Buttons: Back, Register, Update



#	CUST_ID	CUST_NAME	PHONE	ADDRESS	EMAIL	AGENCY_ID
1	C0007	Ram	6667866678	Nellore	shanks@gmail.com	A001
2	C0003	Meenakshi	6667877765	Ranipet	meena@gmail.com	A003
3	C0001	abc	6786786669	Chennai	abc@gmail.com	A001
4	C0002	lmn	4343456756	vellore	lmn@gmail.com	A001
5	C0005	Hema	7776555567	47, Nehrr St,Salem	hema@gmail.com	A001
6	C0017	Raja	7779999987	Vellore	raja@gmail.com	A001
7	C0004	Priya	7897897890	Vellore	shanks@gmail.com	A001
8	C0006	Shreya	5675675678	Chennai	shreya@gmail.com	A003


—
□
×

Gas Supply Management System

Customer Portal

ID:

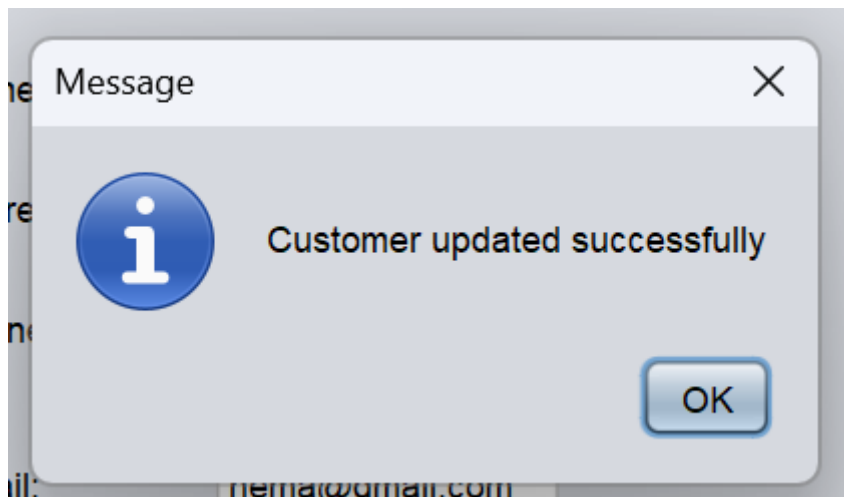
Name:

Address:

Phone:

Email:

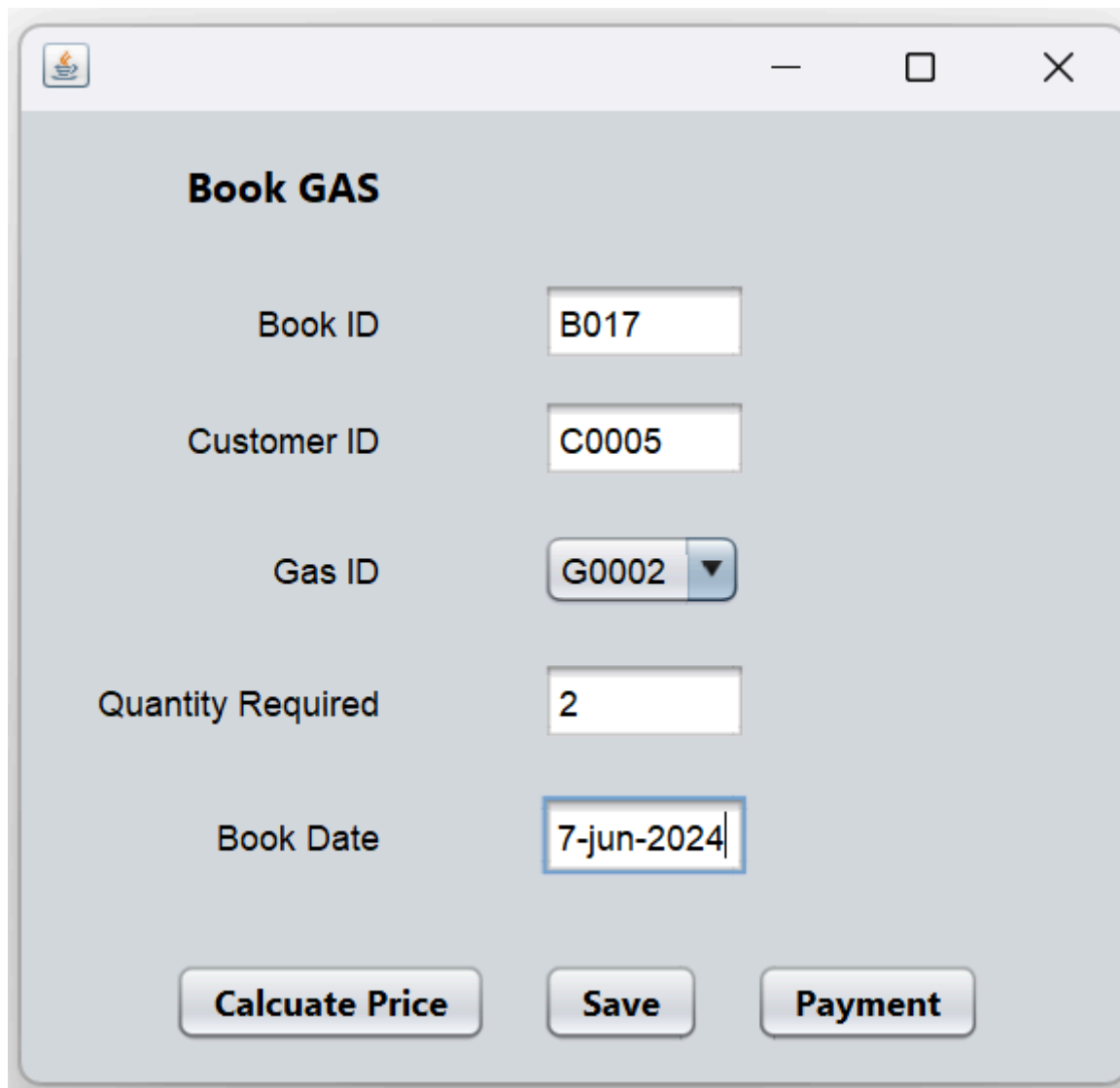
Agency ID ▼



SELECT * FROM "SYSTEM".CU... X						
Max. rows: 100 Fetched Rows: 8 Matching Rows:						
#	CUST_ID	CUST_NAME	PHONE	ADDRESS	EMAIL	AGENCY_ID
1	C0007	Ram	6667866678	Nellore	shanks@gmail.com	A001
2	C0003	Meenakshi	6667877765	Ranipet	meena@gmail.com	A003
3	C0001	abc	6786786669	Chennai	abc@gmail.com	A001
4	C0002	Imn	4343456756	vellore	lmn@gmail.com	A001
5	C0005	Hema	8768768765	Vellore	hema@gmail.com	A001
6	C0017	Raja	7779999987	Vellore	raja@gmail.com	A001
7	C0004	Priya	7897897890	Vellore	shanks@gmail.com	A001
8	C0006	Shreya	5675675678	Chennai	shreya@gmail.com	A003

***Updated address and mobile number for the customer id C0005**

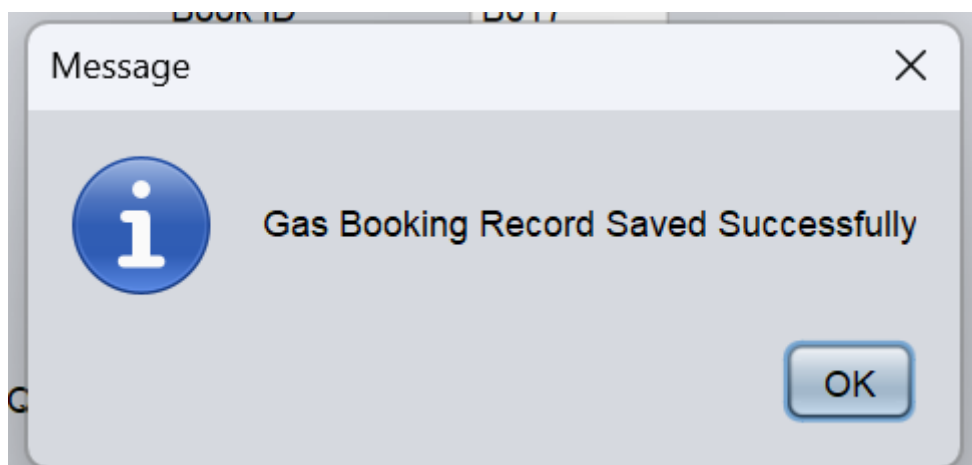
Gas Booking:



The 'Book GAS' window is a light gray dialog box with a standard Windows title bar (minimize, maximize, close buttons). It contains a small icon of a gas burner in the top-left corner. The window is titled 'Book GAS' in bold black text. It features five input fields arranged vertically, each with a label to its left: 'Book ID' with value 'B017', 'Customer ID' with value 'C0005', 'Gas ID' with value 'G0002' and a dropdown arrow, 'Quantity Required' with value '2', and 'Book Date' with value '7-jun-2024'. At the bottom, there are three buttons: 'Calcuate Price' (note the typo), 'Save', and 'Payment'.

Field	Value
Book ID	B017
Customer ID	C0005
Gas ID	G0002
Quantity Required	2
Book Date	7-jun-2024

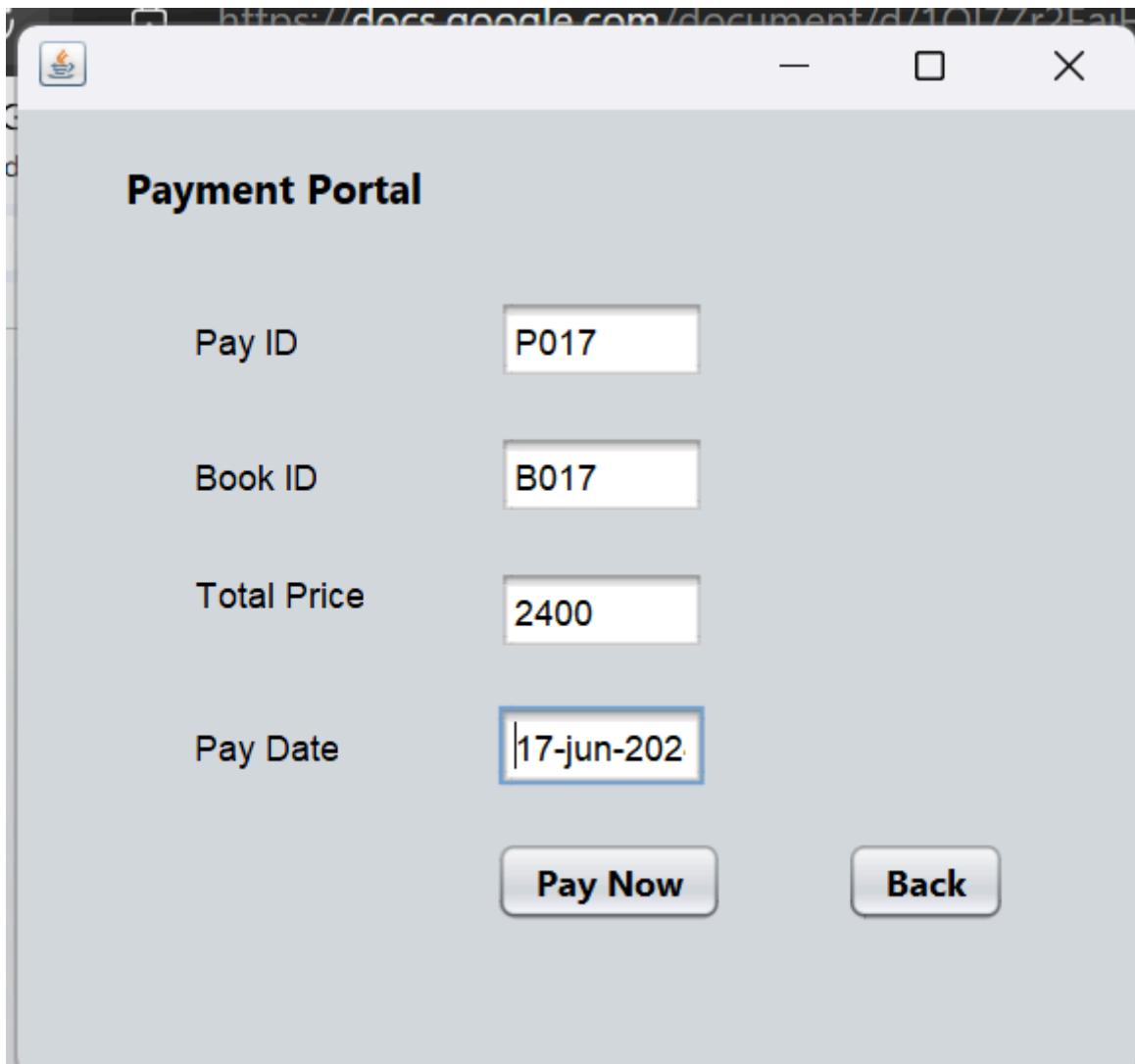
Buttons: **Calcuate Price**, **Save**, **Payment**



Checking inventory before payment:

SELECT * FROM "SYSTEM".IN... ×				
      Max. rows: 100 Fetched Rows: 15				
#	INVENTORY_ID	AGENCY_ID	GAS_ID	QTY_AVAIL
1	I001	A001	G0001	9
2	I002	A001	G0002	10
3	I003	A001	G0003	17
4	I004	A002	G0001	10
5	I005	A002	G0002	10
6	I006	A002	G0003	23
7	I007	A003	G0001	10
8	I008	A003	G0002	10
9	I009	A003	G0003	29
10	I010	A004	G0001	13
11	I011	A004	G0002	10
12	I012	A004	G0003	34
13	I013	A005	G0001	8
14	I014	A005	G0002	10
15	I015	A005	G0003	29

https://docs.google.com/document/d/1O17/r25a1



The image shows a web browser window titled "Payment Portal". It contains four input fields for payment information: "Pay ID" with the value "P017", "Book ID" with the value "B017", "Total Price" with the value "2400", and "Pay Date" with the value "17-jun-202". The "Pay Date" field is currently selected. Below the input fields are two buttons: "Pay Now" and "Back".

Payment Portal

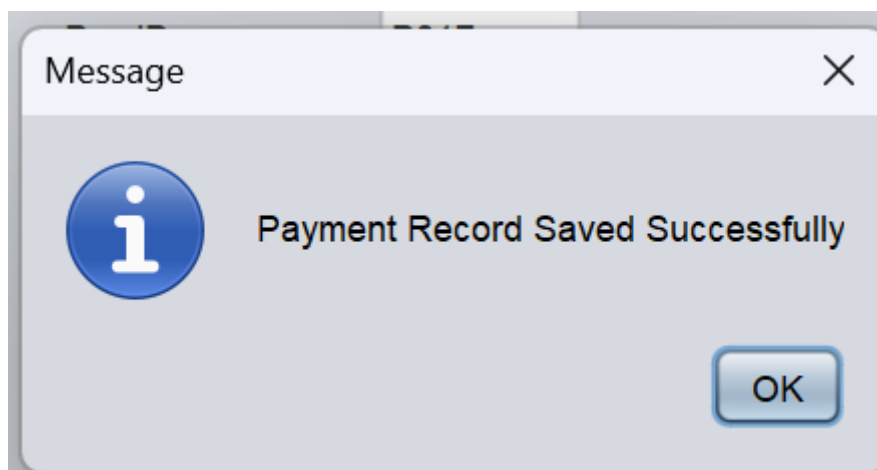
Pay ID P017

Book ID B017

Total Price 2400

Pay Date 17-jun-202

Pay Now Back



Total amount calculation:

```
SQL> select * from total_Amount;
```

BOOK	TOTAL_AMOUNT
----	-----
B014	2000
B015	1000
B016	1000
B017	2400
B018	3600
B019	3600
B020	3600

7 rows selected.

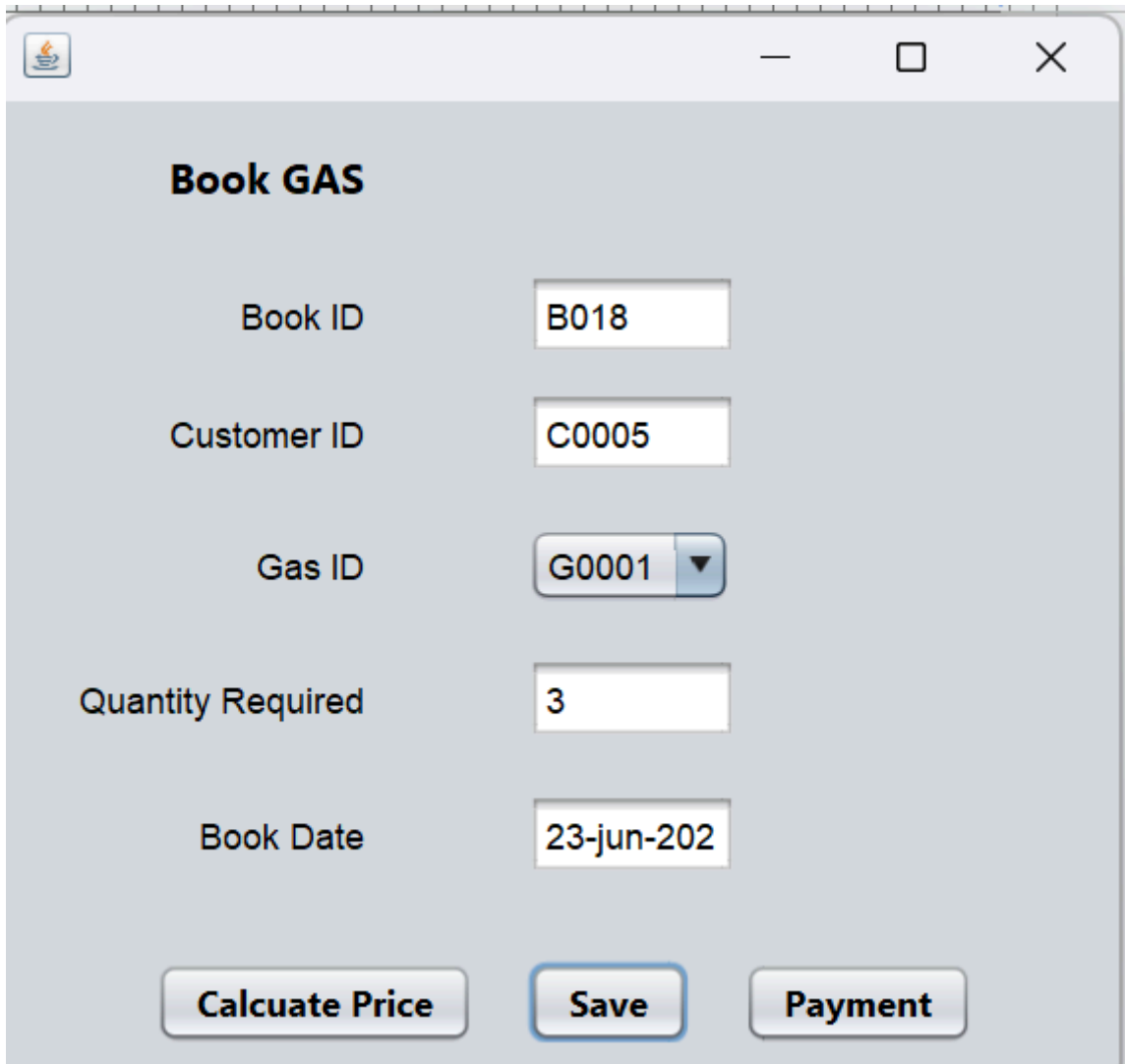
***Here, Check the book id 'B017'.**

Now, Checking the inventory table again.

SELECT * FROM "SYSTEM".IN... ×				
       Max. rows: 100 Fetched Rows: 15				
#	INVENTORY_ID	AGENCY_ID	GAS_ID	QTY_AVAIL
1	I001	A001	G0001	9
2	I002	A001	G0002	8
3	I003	A001	G0003	17
4	I004	A002	G0001	10
5	I005	A002	G0002	10
6	I006	A002	G0003	23
7	I007	A003	G0001	10
8	I008	A003	G0002	10
9	I009	A003	G0003	29
10	I010	A004	G0001	13
11	I011	A004	G0002	10
12	I012	A004	G0003	34
13	I013	A005	G0001	8
14	I014	A005	G0002	10
15	I015	A005	G0003	29

*** It got reduced from the inventory table for the respective gas id and agency id.**

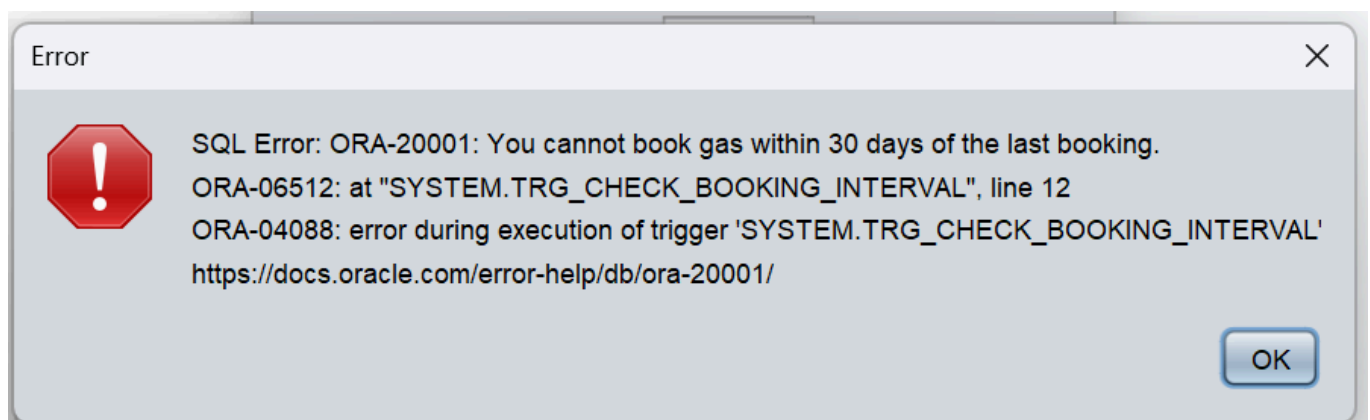
Checking the Interval Days



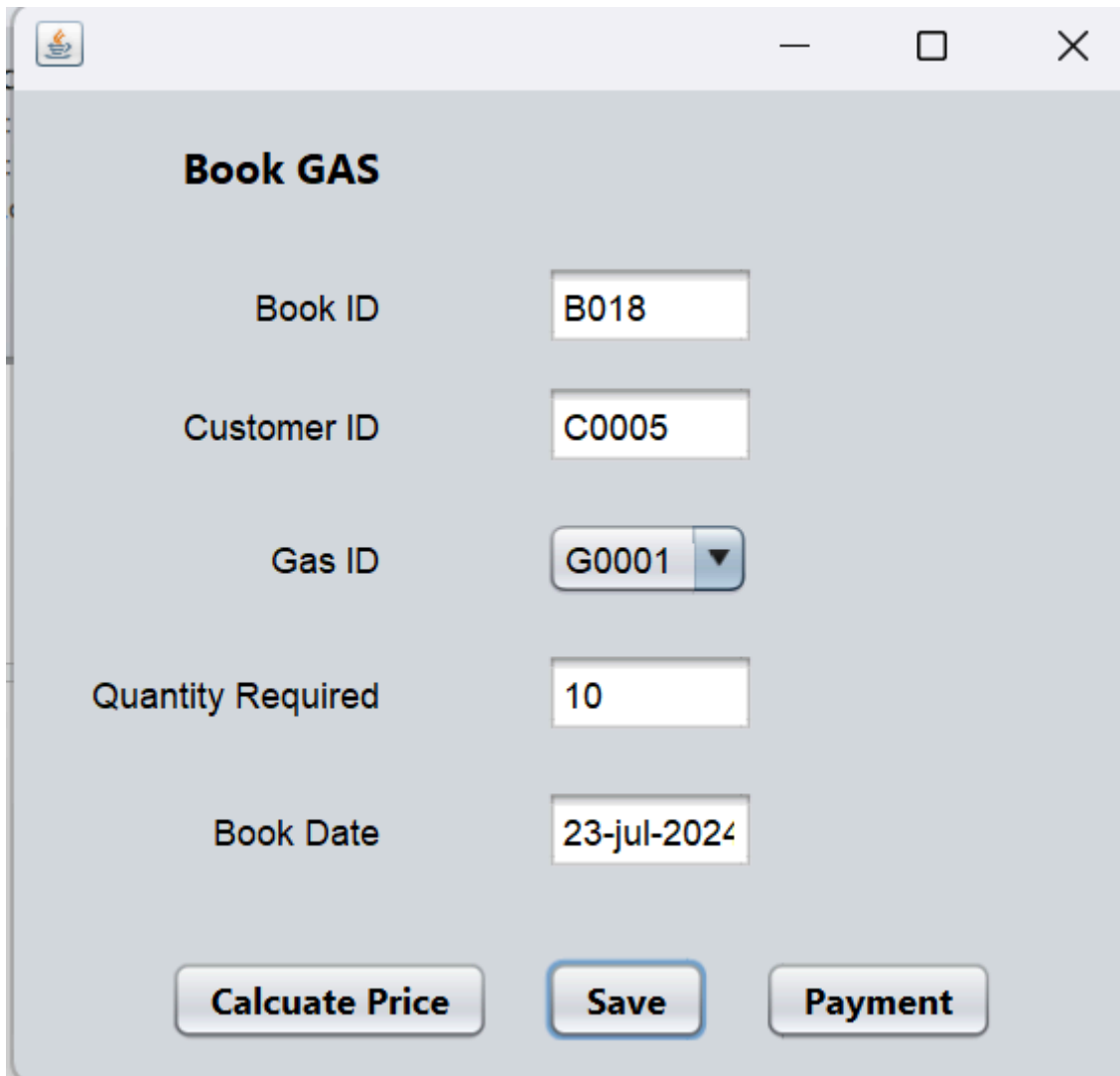
A screenshot of a web application window titled "Book GAS". The window has a standard title bar with a minimize icon, a maximize icon, and a close icon. The main content area is light gray and contains several input fields and buttons. The fields are labeled "Book ID", "Customer ID", "Gas ID", "Quantity Required", and "Book Date". The "Book ID" field contains "B018", "Customer ID" contains "C0005", "Gas ID" is a dropdown menu showing "G0001", "Quantity Required" contains "3", and "Book Date" contains "23-jun-202". At the bottom of the form are three buttons: "Calculate Price", "Save", and "Payment". The "Save" button is highlighted with a blue border.

Book ID	B018
Customer ID	C0005
Gas ID	G0001 ▼
Quantity Required	3
Book Date	23-jun-202

Calculate Price **Save** **Payment**



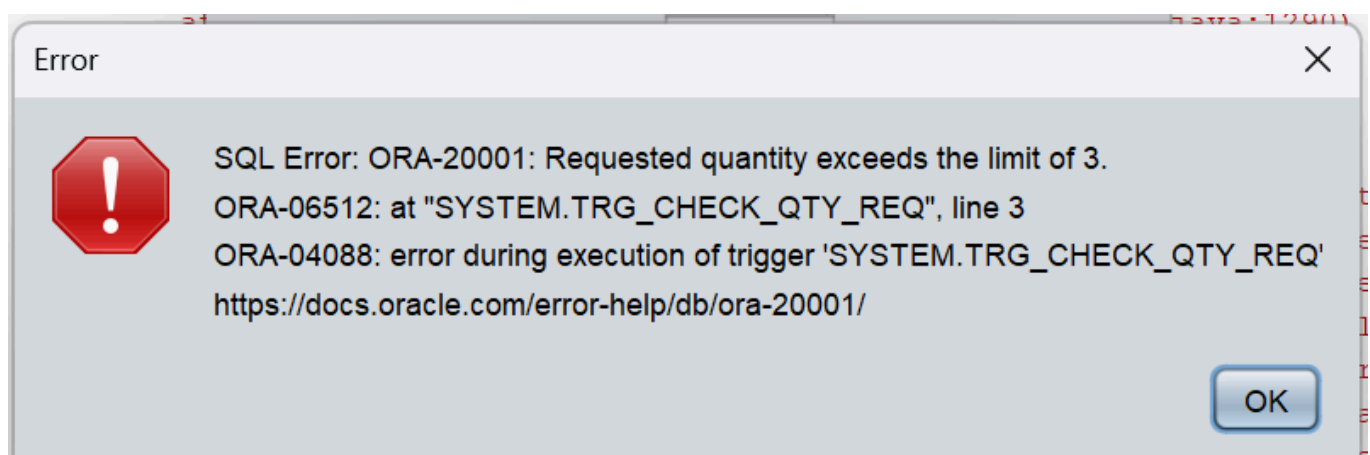
Checking for Quantity..



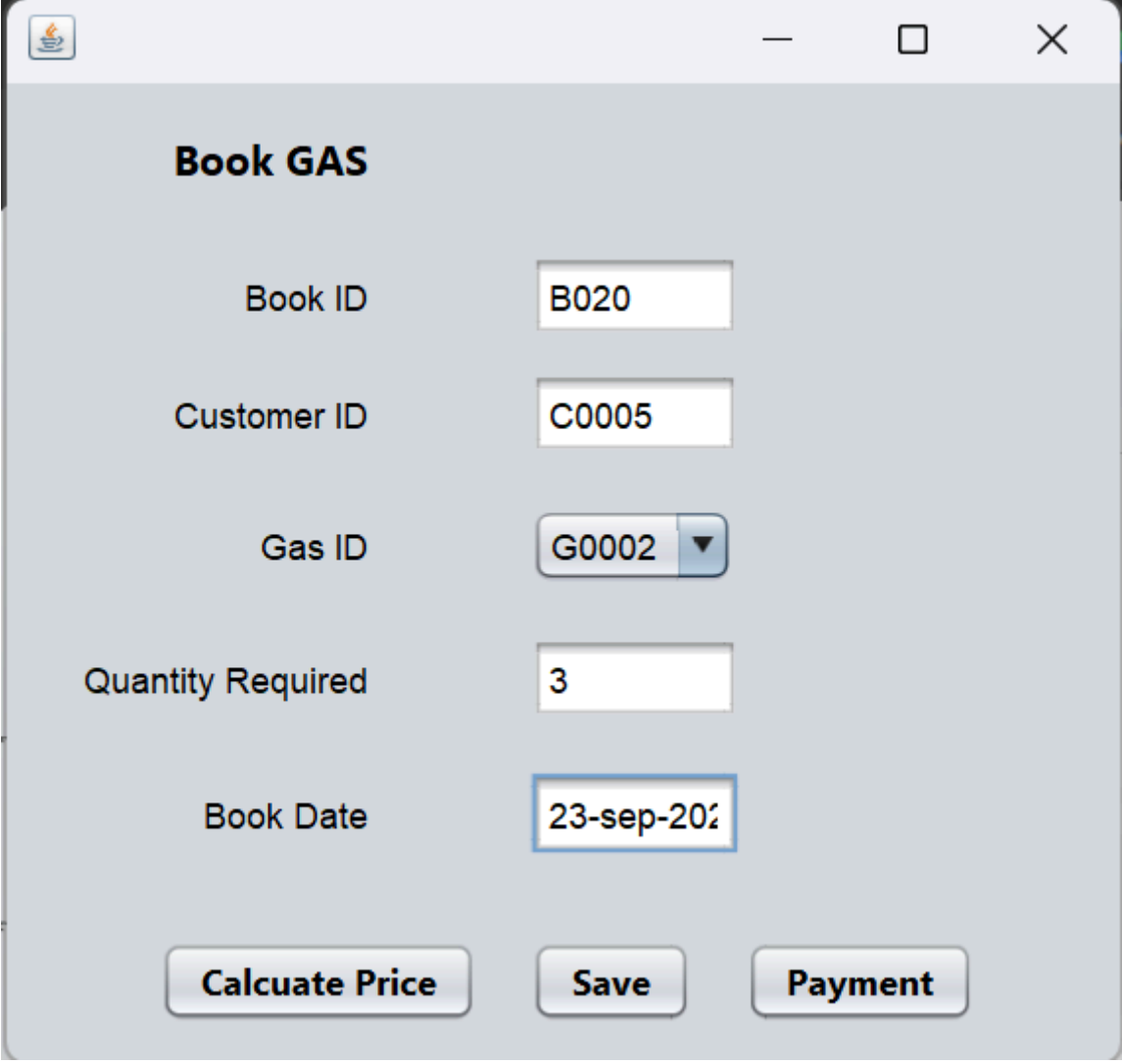
A screenshot of a web application window titled "Book GAS". The window has a standard title bar with a minimize button, a maximize button, and a close button. The form contains several input fields and three buttons at the bottom.

Field	Value
Book ID	B018
Customer ID	C0005
Gas ID	G0001
Quantity Required	10
Book Date	23-jul-2024

Buttons: **Calculate Price**, **Save**, **Payment**



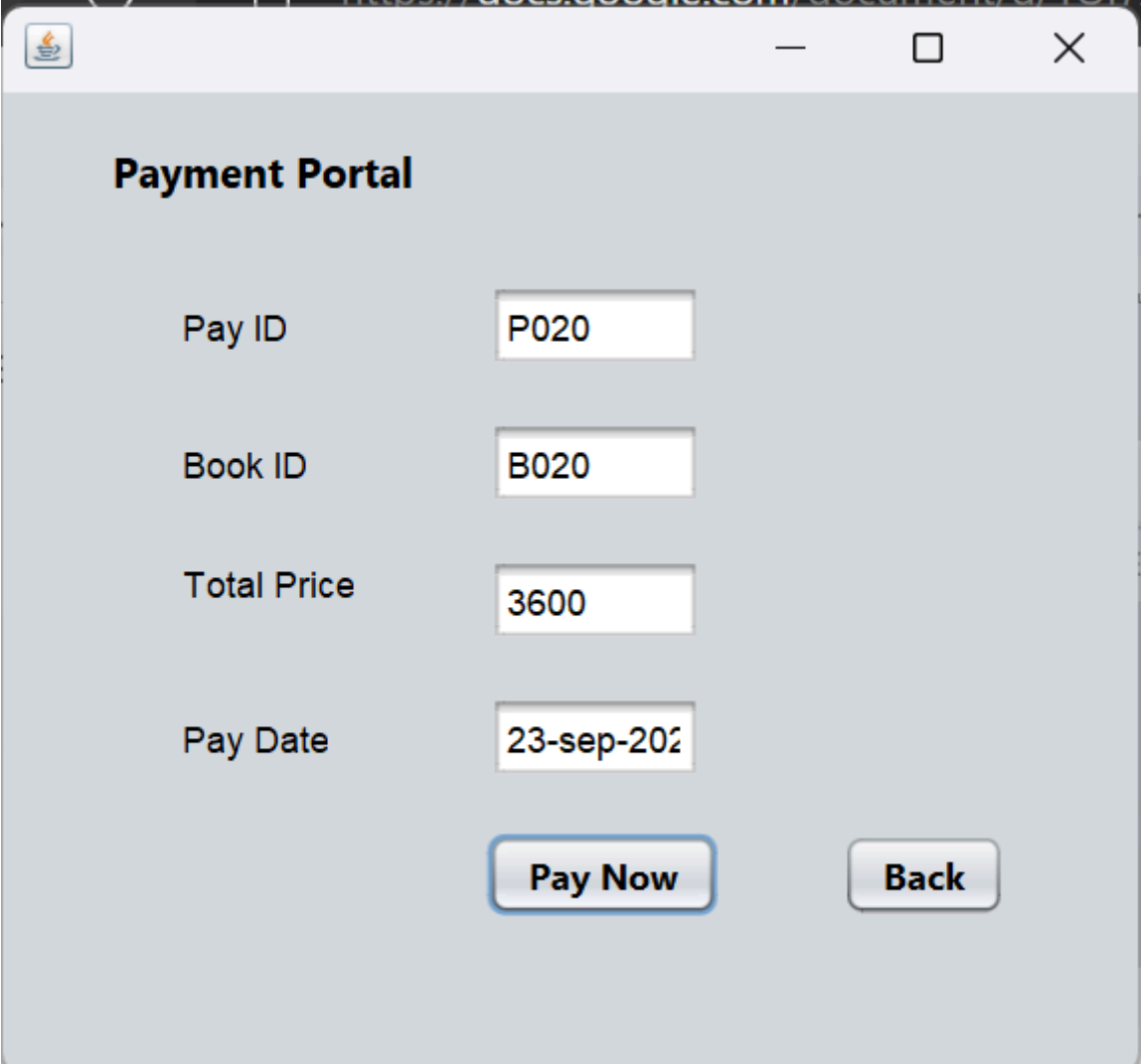
When the req. Quantity is greater than available , gas won't be booked and it throws an error message.



The image shows a software window titled "Book GAS". It contains five input fields and three buttons. The fields are labeled "Book ID", "Customer ID", "Gas ID", "Quantity Required", and "Book Date". The values entered are "B020", "C0005", "G0002", "3", and "23-sep-2021" respectively. The "Gas ID" field is a dropdown menu. At the bottom, there are three buttons: "Calculate Price", "Save", and "Payment".

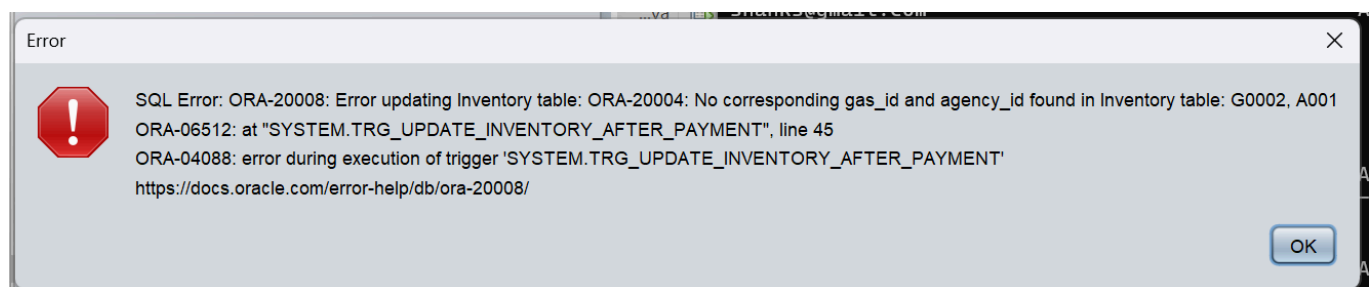
Field	Value
Book ID	B020
Customer ID	C0005
Gas ID	G0002
Quantity Required	3
Book Date	23-sep-2021

Buttons: Calculate Price, Save, Payment



The image shows a web browser window titled "Payment Portal". It contains four input fields for payment information: "Pay ID" with value "P020", "Book ID" with value "B020", "Total Price" with value "3600", and "Pay Date" with value "23-sep-202". Below the fields are two buttons: "Pay Now" and "Back".

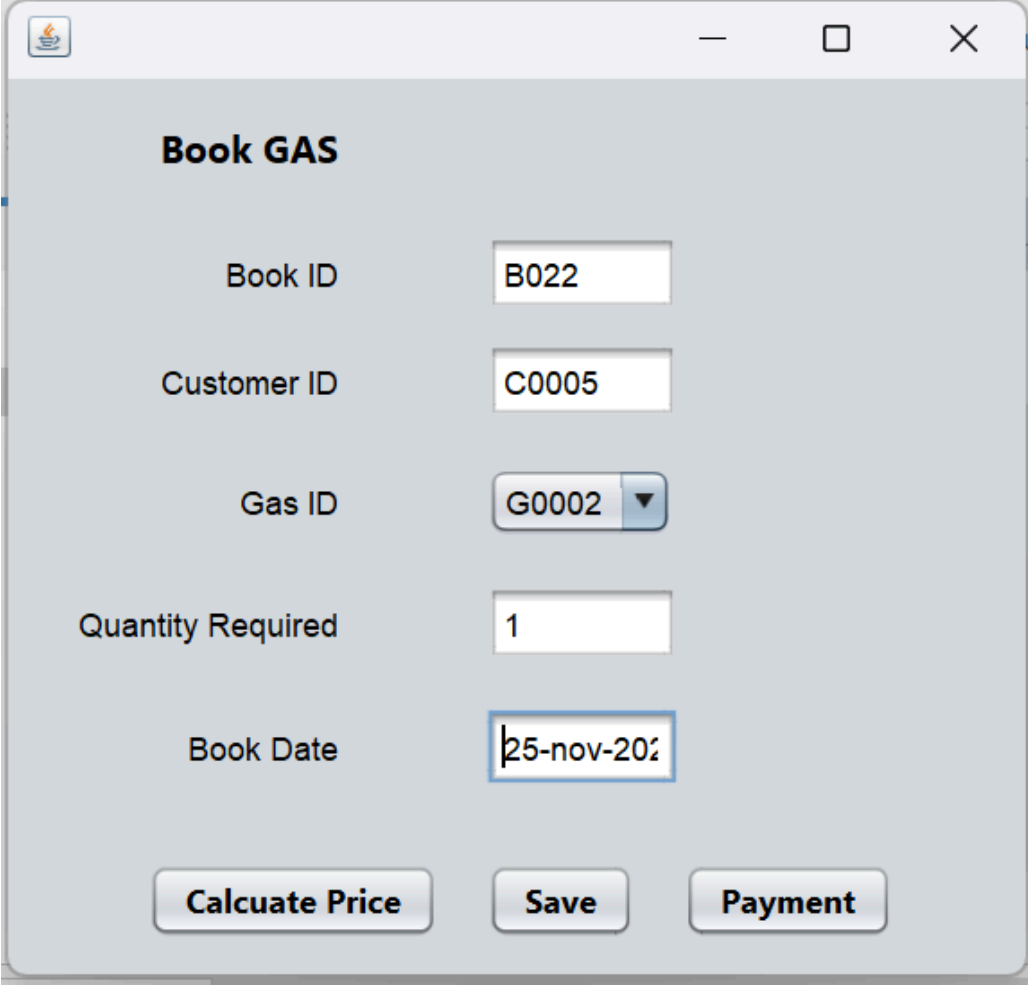
Field	Value
Pay ID	P020
Book ID	B020
Total Price	3600
Pay Date	23-sep-202



SELECT * FROM "SYSTEM".IN... ×				
      Max. rows: 100 Fetched Rows: 15				
#	INVENTORY_ID	AGENCY_ID	GAS_ID	QTY_AVAIL
1	I001	A001	G0001	9
2	I002	A001	G0002	2
3	I003	A001	G0003	17
4	I004	A002	G0001	10
5	I005	A002	G0002	10
6	I006	A002	G0003	23
7	I007	A003	G0001	10
8	I008	A003	G0002	10
9	I009	A003	G0003	29
10	I010	A004	G0001	13
11	I011	A004	G0002	10
12	I012	A004	G0003	34
13	I013	A005	G0001	8
14	I014	A005	G0002	10
15	I015	A005	G0003	29

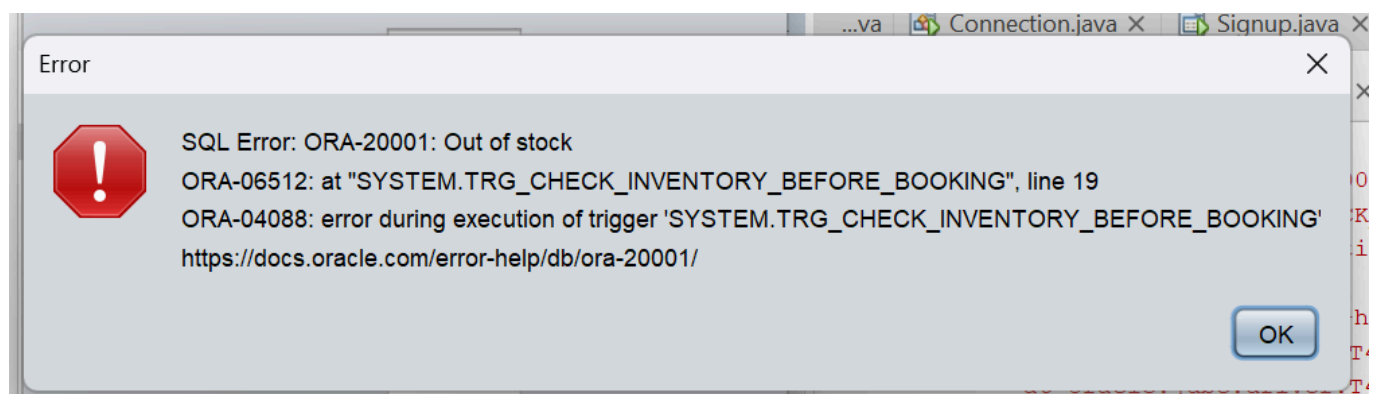
*** In this case, the customer orders 3 quantities of gas. But there are only 2 quantities available for that particular agency id and gas id. Therefore, it throws an error message.**

Checking the inventory:



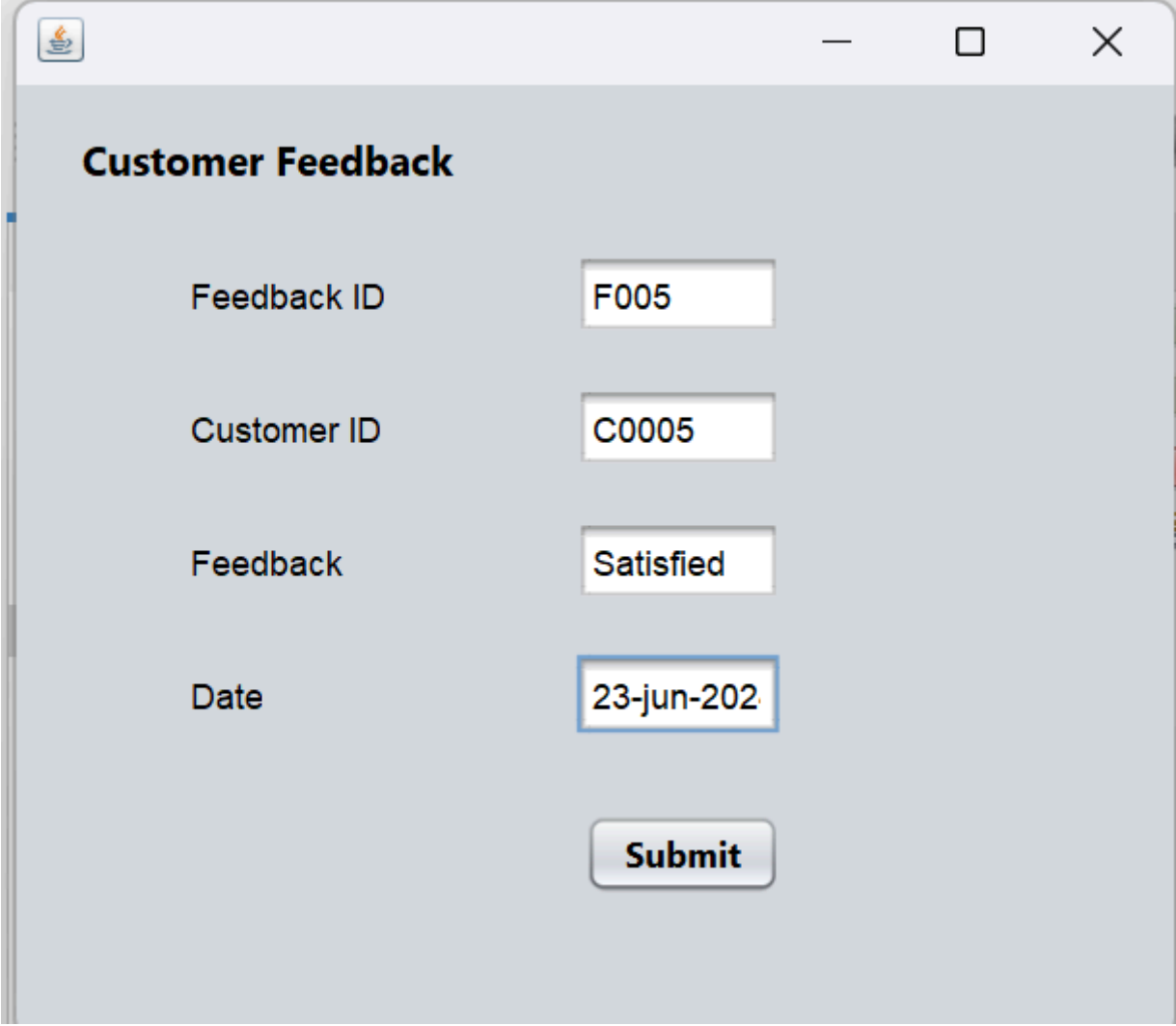
The screenshot shows a Java Swing window titled "Book GAS". It contains several input fields and three buttons at the bottom. The fields are: "Book ID" with value "B022", "Customer ID" with value "C0005", "Gas ID" with a dropdown menu showing "G0002", "Quantity Required" with value "1", and "Book Date" with value "25-nov-2021". The buttons are "Calculate Price", "Save", and "Payment".

Field	Value
Book ID	B022
Customer ID	C0005
Gas ID	G0002
Quantity Required	1
Book Date	25-nov-2021



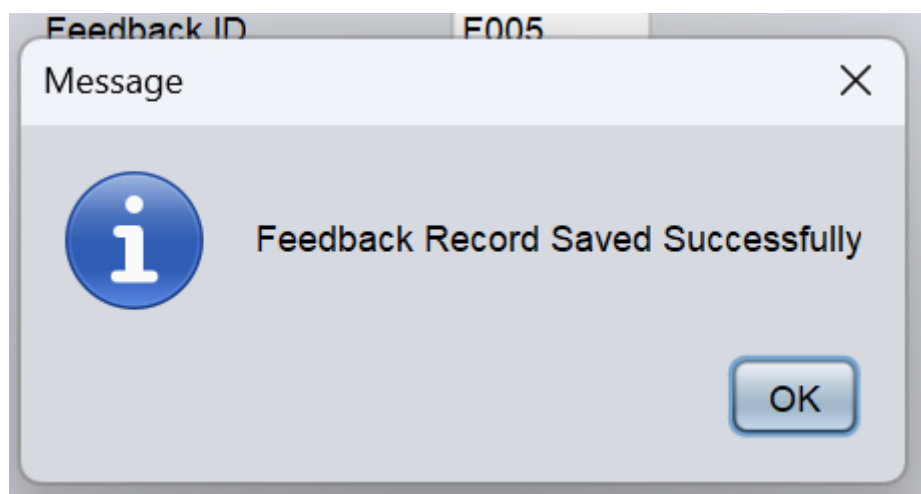
Since the inventory has 0 quantity for the respective gas id and agency id for a book id, it throws an error: Out of Stock.

Feed Back:



A screenshot of a 'Customer Feedback' form window. The window has a title bar with a coffee cup icon and standard minimize, maximize, and close buttons. The form contains four input fields: 'Feedback ID' with the value 'F005', 'Customer ID' with the value 'C0005', 'Feedback' with the value 'Satisfied', and 'Date' with the value '23-jun-202'. A 'Submit' button is located at the bottom center of the form.

Customer Feedback	
Feedback ID	F005
Customer ID	C0005
Feedback	Satisfied
Date	23-jun-202
Submit	



SELECT * FROM "SYSTEM".FE... ×				
      Max. rows: 100 Fetched Rows: 3				
#	FEEDBACK_ID	CUST_ID	FB_CONTENT	FB_DATE
1	F001	C0007	Satisfied	2024-07-...
2	F004	C0003	Good	2024-08-...
▶3	F005	C0005	Satisfied	2024-06-...

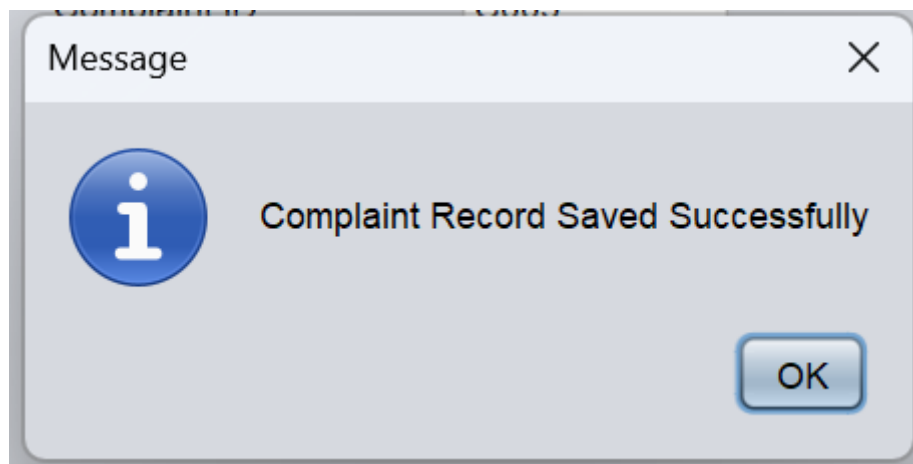
Complaint:


—
□
×

Raise Complaint

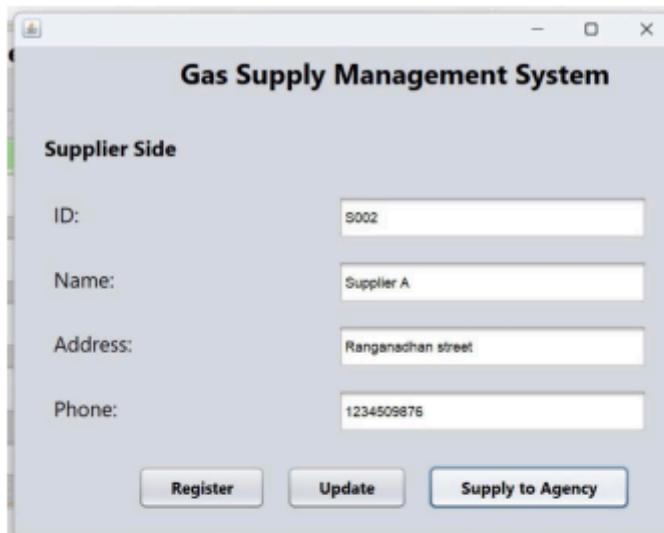
Complaint ID	C005
Customer ID	C0005
Agency ID	A001
Book ID	B017
Complaint	Delay

Submit



SELECT * FROM "SYSTEM".CO... X					
      Max. rows: 100 Fetched Rows: 2					
#	COMP_ID	CUST_ID	AGENCY_ID	BOOK_ID	CONTENT
1	C001	C0007	A001	B001	Delay
▶2	C005	C0005	A001	B017	Delay

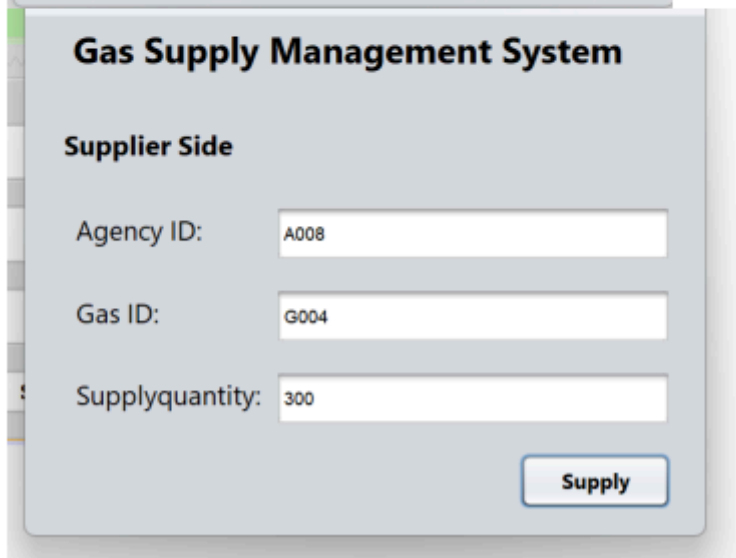
Supplier side:



A screenshot of a software window titled "Gas Supply Management System". Inside, there is a section titled "Supplier Side". It contains four input fields: "ID:" with the value "S002", "Name:" with "Supplier A.", "Address:" with "Ranganathan street", and "Phone:" with "1234509876". Below these fields are three buttons: "Register", "Update", and "Supply to Agency".

Field	Value
ID:	S002
Name:	Supplier A.
Address:	Ranganathan street
Phone:	1234509876

Buttons: Register, Update, Supply to Agency



A second screenshot of the same "Gas Supply Management System" window, showing the "Supplier Side" section. It contains three input fields: "Agency ID:" with the value "A008", "Gas ID:" with "G004", and "Supplyquantity:" with "300". A single "Supply" button is located at the bottom right.

Field	Value
Agency ID:	A008
Gas ID:	G004
Supplyquantity:	300

Button: Supply

Agency side:


—
□
×

Gas Supply Management System

Agency Side:

ID:

Name:

Phone:

Address:

SELECT * FROM "SYSTEM".AG... ×				
     Max. rows: 100 Fetched Rows: 5				
#	AGENCY_ID	AGENCY_NAME	PHONE	ADDRESS
1	A001	A1	1231232123	Chennai
2	A002	A2	4443255543	Ranipet
3	A003	A3	4443255663	Salem
4	A004	A4	4773255663	Trichy
5	A005	A5	8773255663	Vellore